MSI





CAPS

CIRCLE ANALYTICAL GEOMETRY

GRADE 12

CONCEPT	HOW TO LEARN IT?	RELEVANT FORMULAE AND KEYWORDS
Mid – point	Learn formula for midpoint from textbook, study guides and any material you can get your hands on. Work on different questions from study material apply this formula until you feel confident.	$M\left(\frac{y_1+y_2}{2};\frac{y_1+y_2}{2}\right)$
	You need to understand that the mid-point formula is used to determine the middle point of lines or you may also be given the midpoint co-ordinates and one point from which you have to determine the coordinates of another point.	Middle, midpoint, centre
	Practise question that need you to apply midpoint formula, check answers against memo. Consult the educator if you need any clarity. Do as many examples as you can.	Check the correctness of your attempts against the memos.
	Practising questions from different study materials, use the calculator all the time. Knowledge of numbers, halving, algebra and Euclidean geometry comes in very handy in this section. Make sure to get hold of lots of questions from textbooks, study guides and any study materials. Practice lots of question until you feel confident in this topic. Challenge yourself with complicated looking questions.	Check your answers against memos.
	Mid- Point, generally very simple, this is another area where you can collect easy mark. It needs application of the formula which can easily be transferred from the formula sheet always provided in examination papers normally at the back. Beware that sometimes you may be given variables as co-ordinates and you have to determine the numerical values thereof. Look out for Algebraic variables.	

Gradient	Learn formula for gradient from textbook, study guides and any material you can get your hands on. Work on different questions from study material apply this formula until you feel confident.	$m = \frac{y_2 - y_1}{x_2 - x_1}$ $m = \tan\theta$ θ is the inclination of line.
	You may be required to determine the gradient of one or more lines in order to determine whether lines are parallel or perpendicular or to determine the angle of inclination with the x-axis.	Gradient, slope, parallel, perpendicular Parallel lines: $m_1 = m_2$ Perpendicular line: $m_1 \times m_2 = -1$
	Practise question that need you to apply gradient formula, check answers against memo. Consult the educator if you need any clarity. Do as many examples as you can.	Check the correctness of your attempts against the memos.
	Make sure you understand the definition of inclination, so that you don't get confused with which angle they refer to. Apply your knowledge in various questions.	
	Gradient- integrated topics here can be Algebra and Euclidian Geometry, calculating gradient can lead to conclusion of properties of shapes. Theorem of Pythagoras.	
Distance	Learn formula for distance from textbook, study guides and any material you can get your hands on, work on different questions from study material apply this formula until you feel confident. Remember you are allowed to leave your answers in surd form.	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
	You may be required to calculate distance formulae of different lines in order to determine the length of sides to easily classify shapes. For examples the question may say **'prove that ABC is a scalene**	Length, magnitude, distance

	triangle'. This would require that you calculate the length of sides of shape AB, BC, and BA to get and find three different lengths using distance formula.	
	Practise question that need you to apply distance formula, check answers against memo. Consult the educator if you need any clarity.	Check the correctness of your attempts against the memos.
	Practising questions from different study materials, use the calculator all the time. Knowledge of numbers, halving, algebra and Euclidean geometry comes in very handy in this section. Make sure to get hold of lots of questions from textbooks, study guides and any study materials. Practice lots of question until you feel confident in this topic. Challenge yourself with complicated looking questions.	Check your answers against memos. Make sure if the question wants the answer in surd form.
	Distance Formula- integrated topics here can be Algebra and Euclidian Geometry, calculating lengths of sides can lead to conclusion of properties of shapes. For example, prove that a triangle is isosceles, scalene, equilateral using the distance formula.	
Inclination	Learn formula for angle of inclination from textbook, study guides and any material you can get your hands on. Work on different questions from study material apply this formula until you feel confident, remember that if you get a negative angle, you need to subtract it from 180 degrees to get the correct angle of inclination. **Definition of inclination**: It is the angle formed between the positive x – axis and a line, measured in an anti-clockwise direction.	$\theta = \tan^{-1}(m)$ $m = \tan\theta$ θ is the inclination of line.
	You may need practice a lot of questions that need you to calculate angles of inclination. Use questions from textbooks, study guides and any kind of	Angle given

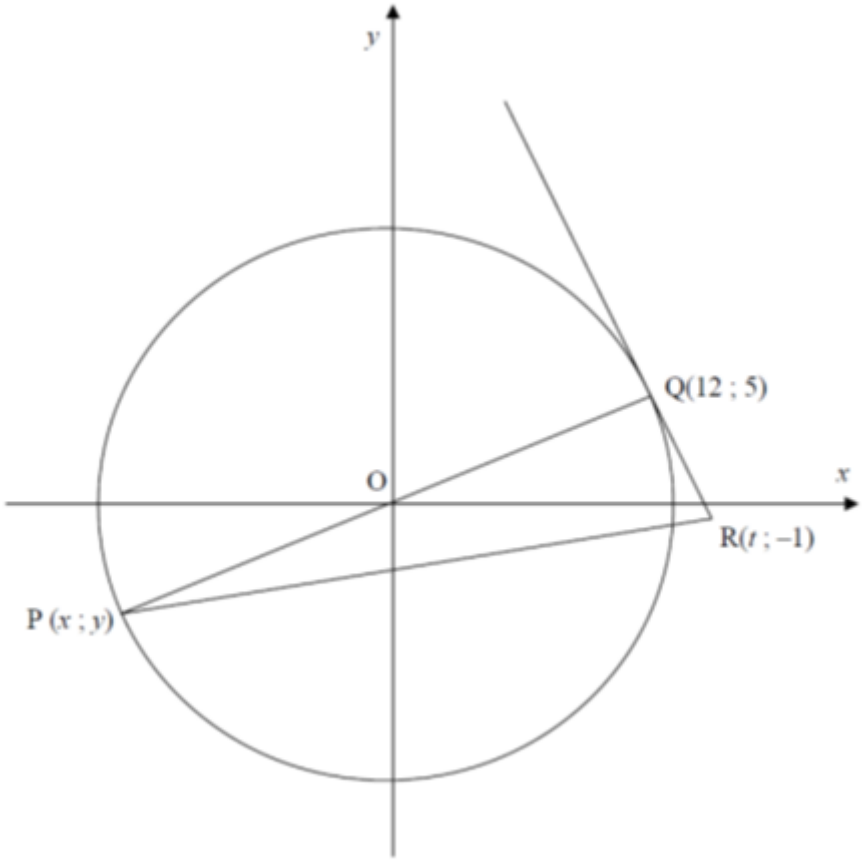
	material you can lay your hands on.	
	Practise question that need you to apply inclination angle formula, check answers against memo. Consult the educator if you need any clarity. Do as many examples as you can.	Check the correctness of your attempts against the memos.
	Practising questions from different study materials, use the calculator all the time. Knowledge of numbers, halving algebra and Euclidean geometry comes in very handy in this section. Make sure to get hold of lots of questions from textbooks, study guides and any study materials. Practice lots of questions until you feel confident in this topic. Challenge yourself with complicated looking questions.	Check the correctness of your attempts against the memos.
	Inclination – Integrated topics are trigonometry, geometry and algebra.	
Equation of circle Mid – point $(a\,;b)$	Learn equation of a circle from textbooks, study guides and any material you can get your hands on. Work on different questions from study material apply this formula until you feel confident. Remember that the radius needs to be squared.	$x^2+y^2=r^2$ with centre $(0\,;0)$ $(x-a)^2+(y-b)^2=r^2$ with centre $(a\,;b)$
	Practice a lot of questions with circle that has centre (0; 0) and radius 'r'. Practice circle that has transformed and shifted from the origin to any point on the set of axes to centre $(a;\ b)$, remember that $(a;\ b)$ can have co-ordinates (0; 0).	Determine the equation of a circle. Determine the coordinate of s centre, midpoint Determine the length of the radius
	Practise question that need you to apply midpoint formula, check answers against memo. Consult the educator if you need any clarity. Do as many examples as you can.	Check the correctness of your attempts against the memos.

	Practising questions from different study materials, use the calculator all the time. Knowledge of numbers, halving algebra and Euclidean geometry comes in very handy in this section. Make sure to get hold of lots of questions from textbooks, study guides and any study materials. Practice lots of questions until you feel confident in this topic. Challenge yourself with complicated looking questions.	Check the correctness of your attempts against the memos.
	Equation of a circle with centre $(a; b)$ – Geometry and algebra are integrated topics in this area.	
Equation of tangent to a circle	Learn formula for equation of the tangent to the circle from textbook, study guides and any material you can get your hands on. Work on different questions from study material apply this formula until you feel confident. Remember the theorem: The tangent is perpendicular to the radius	$y = mx + c$ $y = ax + q$ $y - y_1 = m(x - x_1)$ $m_1 \times m_2 = -1$
	Practice a lot a question on finding the equation of the tangent to the circle. Remember that this of this line will always be perpendicular to the radius at point of contact with the circle	Tangent, equation of a line passing through point ... on the circumference.
	Practise question that need you to apply equation of the tangent formula, check answers against memo. Consult the educator if you need any clarity. Do as many examples as you can.	Check the correctness of your attempts against the memos.
	Practising questions from different study materials, use the calculator all the time. Knowledge of numbers,	Check the correctness of your attempts against the memos.

	halving algebra and Euclidean geometry comes in very handy in this section. Make sure to get hold of lots of questions from textbooks, study guides and any study materials. Practice lots of questions until you feel confident in this topic. Challenge yourself with complicated looking questions.	
	Equation of a tangent to the circle – the integrated topics in this one is algebra, functions and geometry.	

Activity 1

O is the centre of the circle in the figure below. P$(x; y)$ and Q$(12; 5)$ are two points on the circle. POQ is a straight line. The point R$(t; -1)$ lies on the tangent to the circle at Q.



1.1 Determine the equation of the circle. (3)

1.2 Determine the equation of the straight line through P and Q. (2)

1.3 Determine x and y, the coordinates of P. (2)

1.4 Show that the gradient of QR is $-\frac{12}{5}$. (2)

1.5 Determine the equation of QR in the form $y =$ .... (3)

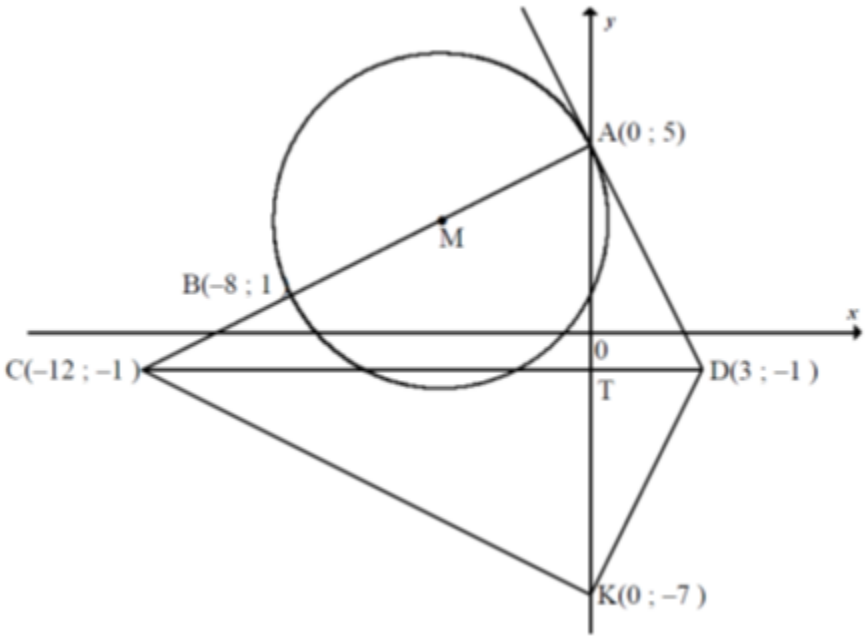
1.6 Calculate the value of t. (2)

1.7 Determine an equation of the circle with centre $Q(12; 5)$ and passing through the origin. (3)

[17]

Activity 2

A$(0; 5)$ and B$(-8; 1)$ are two points on the circumference of the circle with centre M, in a Cartesian plane. M lies on AB. DA is a tangent to the circle at A.
The coordinates of D are $(3; -1)$ and the coordinates of C are $(-12; -1)$.
Points C and D are joined. K is the point $(0; -7)$. CTD is a straight line.



2.1 Show that the coordinates of M, the midpoint of AB, are $(-4; 3)$. (1)

2.2 Determine the equation of the tangent AD. (4)

2.3 Determine the length of AM. (3)

2.4 Determine the equation of a circle centre M in the form $ax^2 + by^2 + cx + dy + e = 0$. (4)

2.5 Quadrilateral ACKD is one of the following:

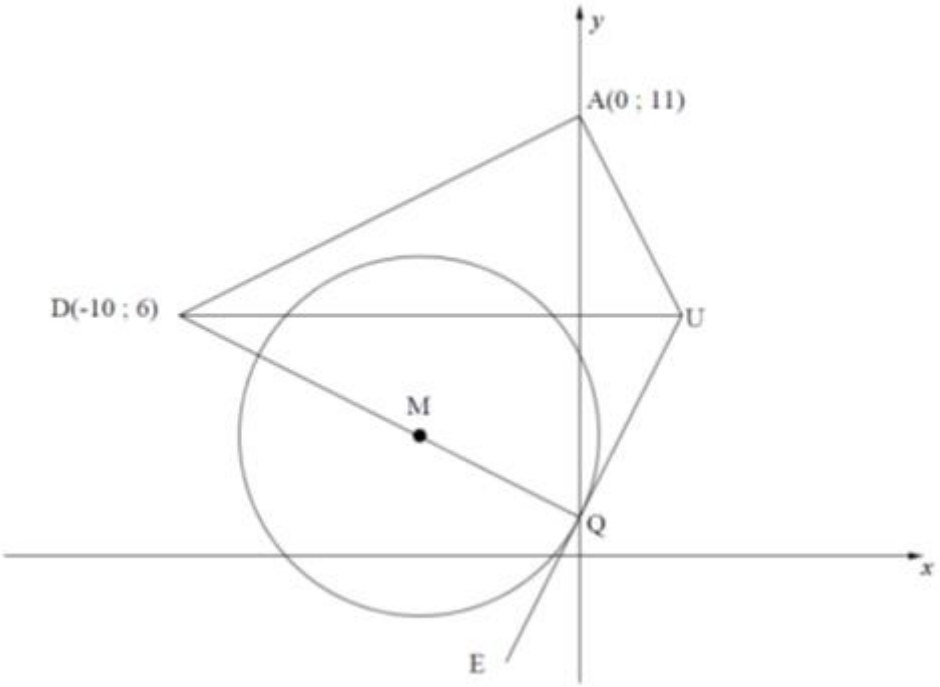
Parallelogram; Kite; Rhombus; Rectangle.

Which one is it? Justify your answer. (4)

[16]

Activity 3

$x^2 + y^2 + 8x - 6y = -5$, is the equation of the circle with centre M. UE is a tangent to the circle at Q. QMD, DA, AU and UQE are straight lines. DU is parallel to the x-axis.



3.1 Determine the coordinates of M, the centre of the circle. (4)

3.2 Calculate the coordinates of Q, if $y > 2$. (3)

3.3 Calculate the equation of tangnet UE. (4)

3.4 Write down the equation of DU. (1)

3.5 Calculate the coordinates of U. (2)

3.6 Prove that QUAD is a cyclic quadrilateral. (6)

[20]

Activity 4

4.1 Determine the centre and radius of the circle with the equation $x^2+y^2 + 8x + 4y - 38 = 0$. (4)

4.2 A second circle has the equation $(x - 4)^2 + (y - 6)^2 = 26$. Calculate the distance between the centres of the two circles. (2)

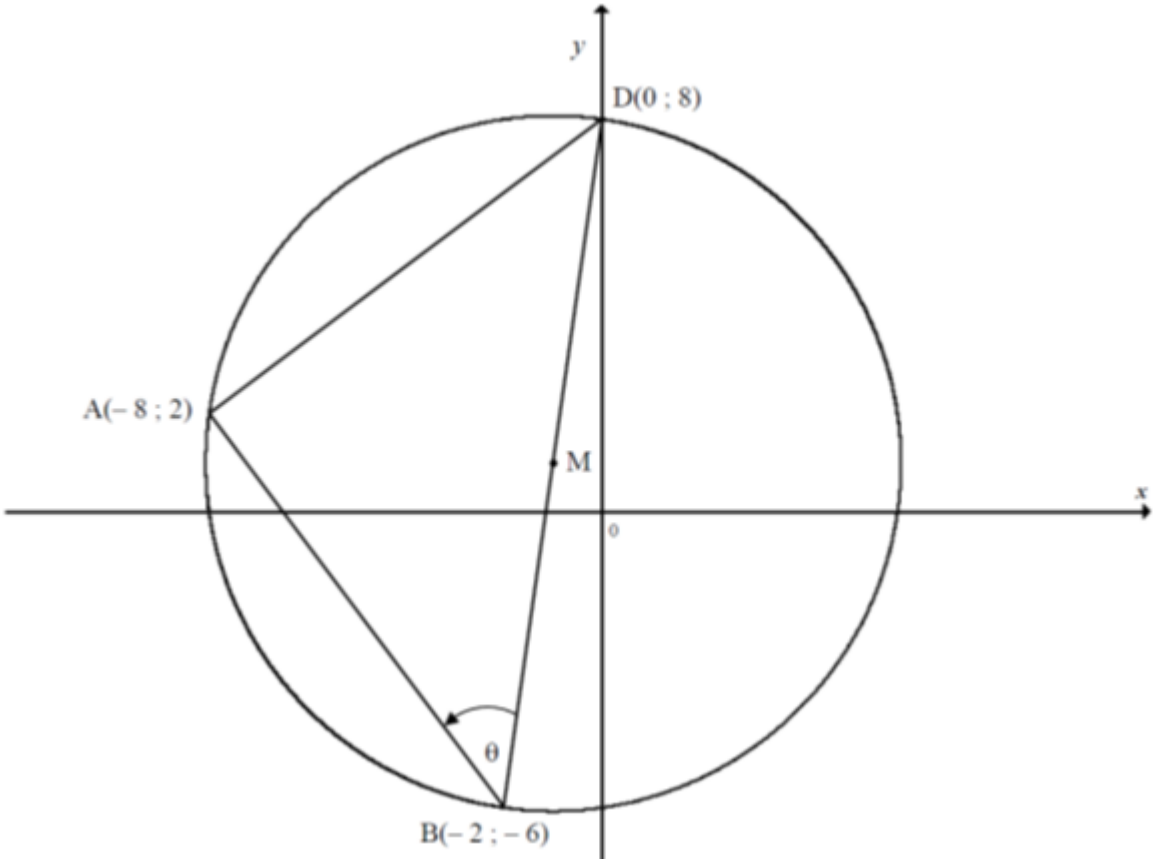
4.3 Hence, show that the circles desribed in Activity 4.1 and 4.2 intersect each other. (3)

4.4 Show that the two circle intersect along the line $y = -x + 4$. (4)

[13]

Activity 5

$A(-8;2)$, $B(-2;-6)$ and $D(0;8)$ are the vertices of a triangle that lies on the circumference of a circle with diameter BD and centre M, as shown in the figure below.



5.1 Calulate the coordinate of M. (2)

5.2 Show that $(-8;2)$ lies on he line $y = 7x + 58$. (1)

5.3 What is the realtionship between the line $y = 7x + 58$ and the circle centred at M? Motivate your answer. (5)

5.4 Calculate the lengths of AD and AB. (4)

5.5 Prove that $D\hat{A}B = 90°$. (3)

5.6 Write down the size of angle θ. (1)

[16]

Activity 6

The equatio of a circle is $x^2 + y^2 - 8x + 6y = 15$.

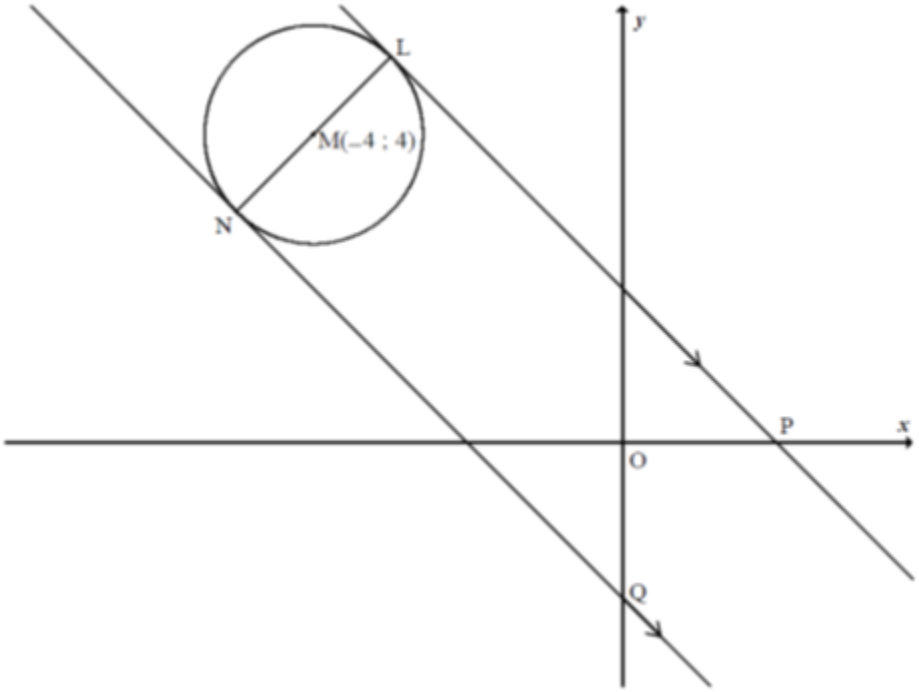
6.1 Prove that the point $(2; -9)$ is on the circumference of the circle. (2)

6.2 Determine an equation of the tangent to the circle at point $(2; -9)$. (7)

[9]

Activity 7

The line LP, with equation $y + x - 2 = 0$, is a tangent at L to the circle with centre $M(-4; 4)$. LN is a diameter of the circle. Also, LPNQ, where P lies on the x-axis, and Q lies on the y-axis.



7.1 Determine the equation of the diameter LN. (3)

7.2 Calculate the coordinates of L. (2)

7.3 Determine the equation of the circle. (3)

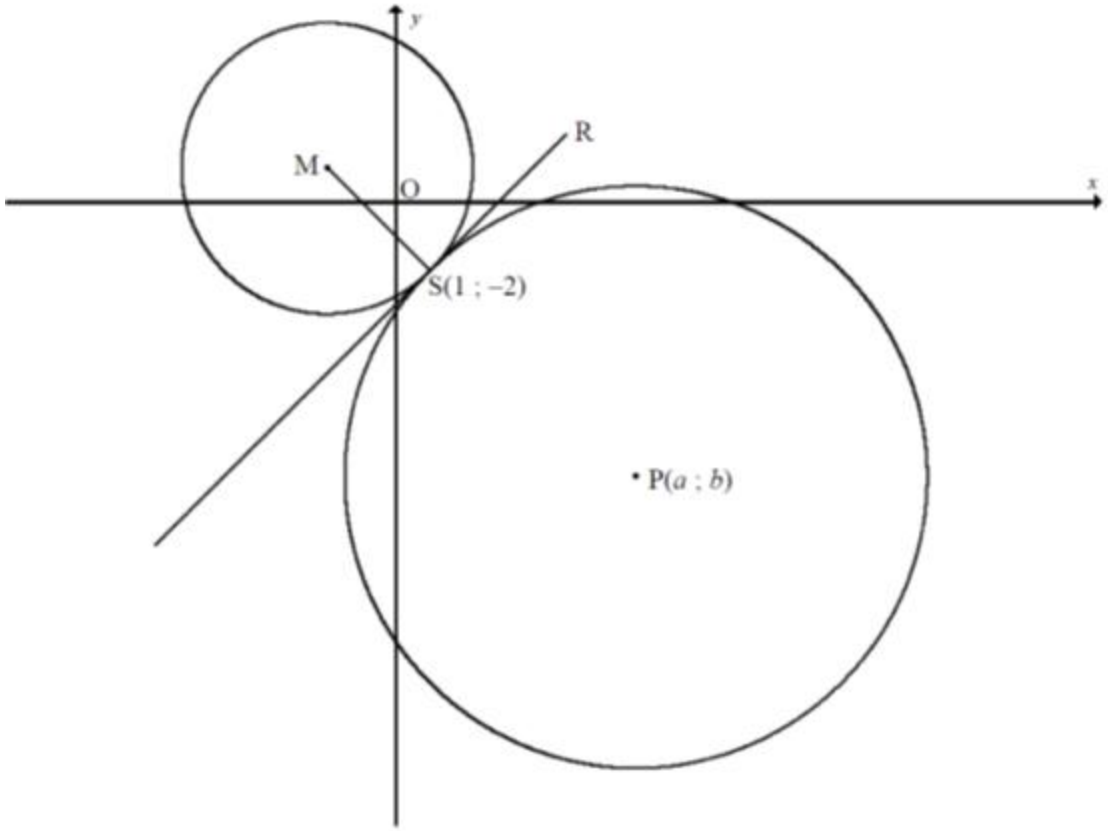
7.4 Write down the coordinates of N. (3)

7.5 Write down the equation of NQ. (3)

[14]

Activity 8

In the figure below, a circle with centre M is drawn. The equation of the circle is $(x+2)^2+(y-1)^2=r^2$. $S(1;-2)$ is a point on the circle. SR is a tangent to the circle.



8.1 Write down the coordinates of M and the radius of the circle centred at M. (4)

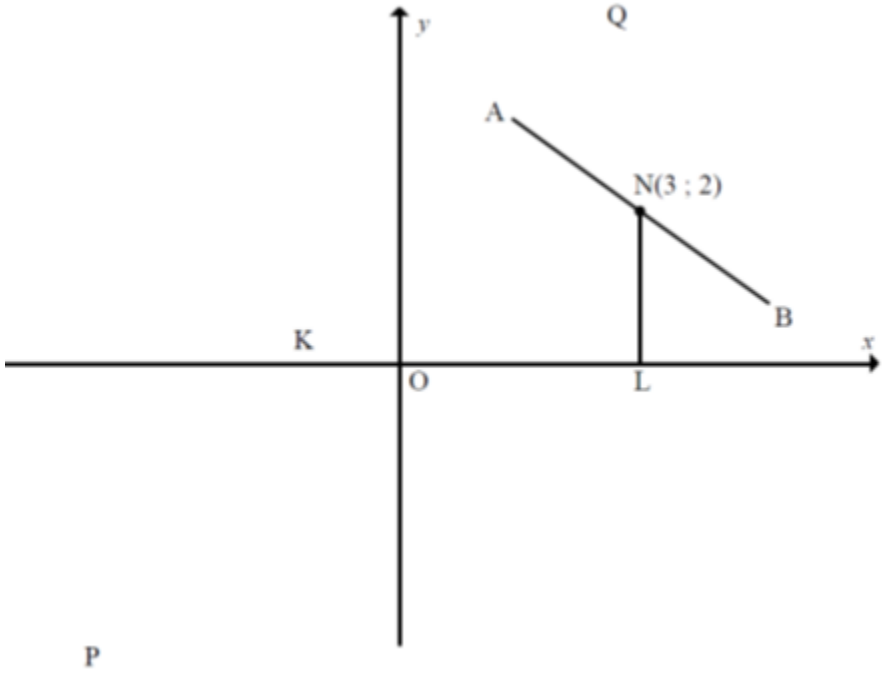
8.2 Determine the equation of the tangent RS in the form $y=mx+c$. (4)

8.3 The circles having centres P and M touch each externally at point S. SR is a tangent to both circles. If MS: MP $= 1:3$, determine the coordinates $(a;b)$ of point P. (8)

[16]

Activity 9

A circle centred at N(3; 2) touches the x-axis at point L. the line PQ, defined by the equation $y = \frac{4}{3}x + \frac{4}{3}$, is a tangent to the same circle at point A.



9.1 Why is NL perpendicular to OL? (1)

9.2 Determine the coordinates of L. (1)

9.3 Determine the equation of the circle with centre N in the form $(x-a)^2 + (y-b)^2 = r^2$. (3)

9.4 Calculate the length of KL. (3)

9.5 Determine he equation of the diameter AB in the form $y = mx + c$. (4)

9.6 Show that the coordinates of A are $\left(\frac{7}{5}; \frac{16}{5}\right)$. (3)

9.7 Calculate the length of KA. (3)

9.8 Why is KLNA a kite? (2)

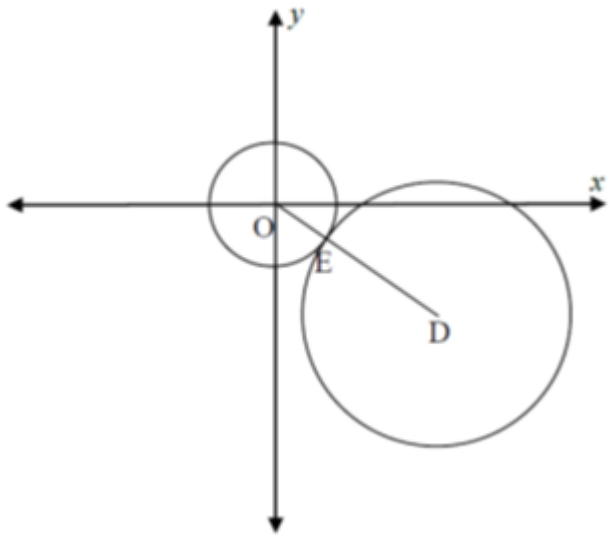
9.9 Show that $A\hat{B}K = 45°$. (3)

9.10 If the given circle is reflected about the x-axis, give the coordinates of the centre of the new circle. (1)

[24]

Activity 10

10.1 The diagram below shows two circles touchng at point $E(2;-1)$.
The smaller circle has its centre at the origin and a bigger circle has centre $D(a;-3)$.
The length of the radius of the bigger circle is twice the length of the radius of the smaller circle.



10.1.1 Calculate the length of the radius of the smaller circle. (2)

10.1.2 Determine the value of a if $a > 0$. (4)

10.1.3 Determine the equation of the bigger circle. (3)

10.1.4 Determine the equation of a tangent to both circles at E. (4)

10.2 The length of the diameter of the circle $x^2 + y^2 - 4x + 5y + k = 0$ is 24.

Determine:

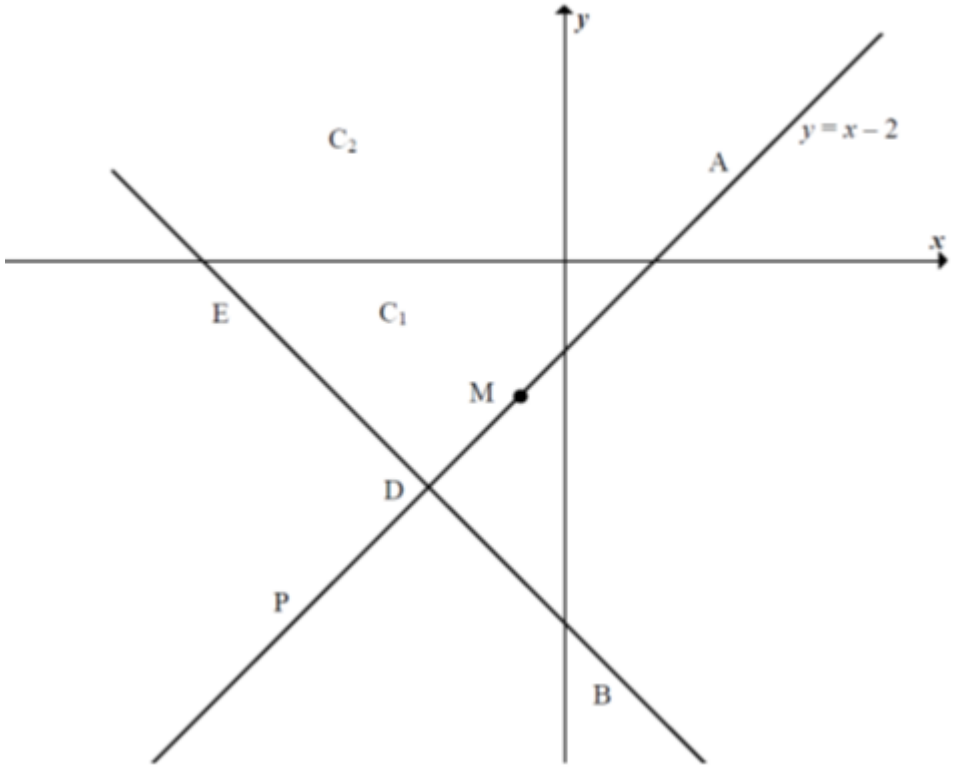
10.2.2 The coordinates of the centre of the circle. (4)

10.2.3 The value of k. (4)

[20]

Activity 11

Circles C_1 and C_2 in the figure below have the same centre M. P is a point on C_2.
PM intersects C_1 at D. the tangent BD to C_1 intersects C_2 at B.
The equation of C_1 is given by $x^2 + 2x + y^2 + 6y + 2 = 0$ and the equation of the line PM is $y = x - 2$.



11.1 Determine the following:

11.1.1 The coordinates of centre M. (3)

11.1.2 The radius of circle C_1. (1)

11.2 Determine the coordinates of D, the point where line PM and circle C_2 intersect. (5)

11.3 If it is given that DB $= 4\sqrt{2}$, determine MB, the radius of circle C_2. (3)

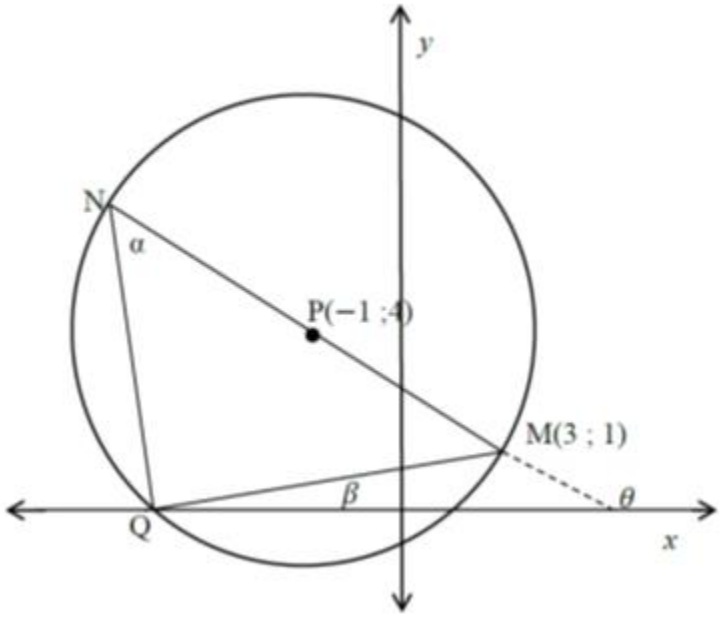
11.4 Write down the equation of C_2 in the form $(x - a)^2 + (y - b)^2 = r^2$. (2)

11.5 Is the point F$(2\sqrt{5}; 0)$ inside circle C_2? Support your answer with calculations. (4)

[18]

Activity 12

In the diagram below, $M(3;1)$, Q and N lie on the circumference of the circle with centre $P(-1;4)$ and form ΔMQN. NPM is a straigh line.



12.1 Determine the equation of the circle. (4)

12.2 Why is $N\hat{Q}M = 90°$? (1)

12.3 Show that the coordinates of Q are $(-4;0)$. (3)

12.4 Calculate the gradient of MN. (2)

12.5 Hence, calculate the size of α. (5)

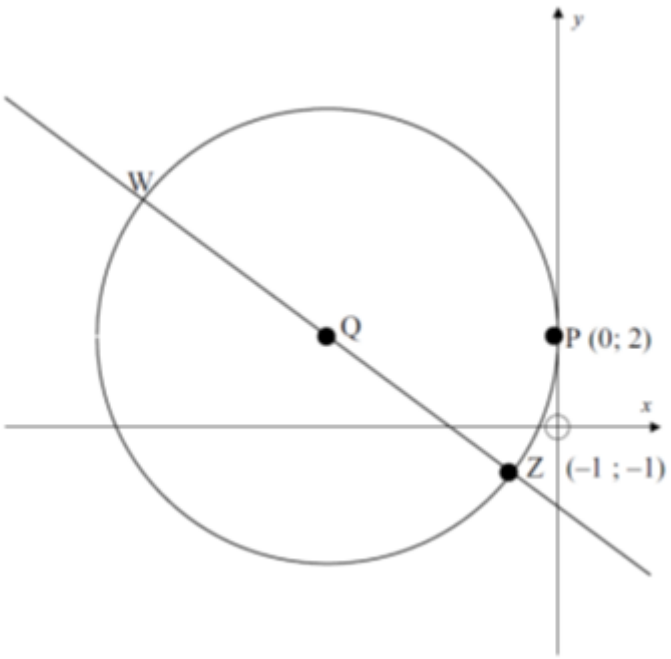
12.6 Determine the equation of a tangent to the circle at M. (5)

[20]

Activity 13

In the diagram below, centre Q of the circle lies on the straight line $3x + 4y + 7 = 0$.

The straight line cuts the circle at W and $Z(-1; -1)$. The circle touches the y-axis at $P(0; 2)$.



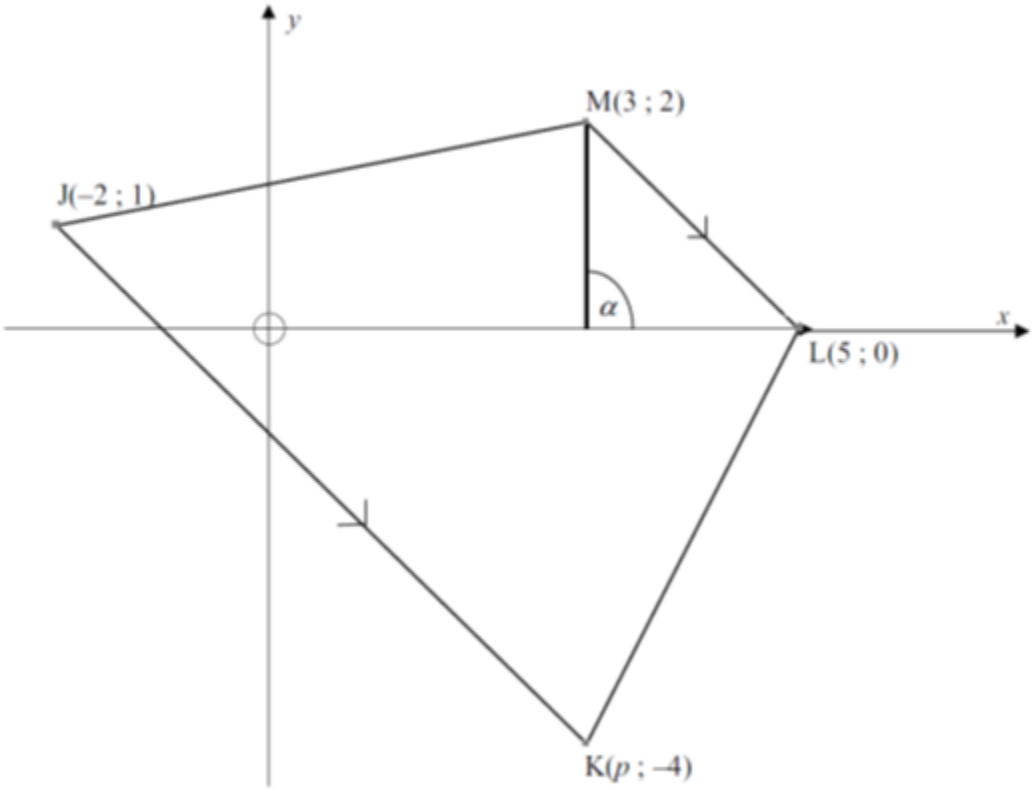
13.1 Determine the equation of the circle in the form $(x - a)^2 + (y - b)^2 = r^2$. (5)

13.2 Determine the length of diameter WZ. (1)

[6]

Activity 14

In the digram below, J$(-2; 1)$, K$(p; -4)$, L$(5; 0)$ and M$(3; 3)$ are the vertices of trapezium JKLM. Also, JK ∥ ML.



14.1 Show that $p = 3$. (4)

14.2 Calculate JK: LM in the simplest form. (5)

14.3 Q$(x; y)$ on JK is such that JQLM is a parallelogram, determine the coordinates of Q. (5)

14.4 Determine the equation of the line passing through K and M. (2)

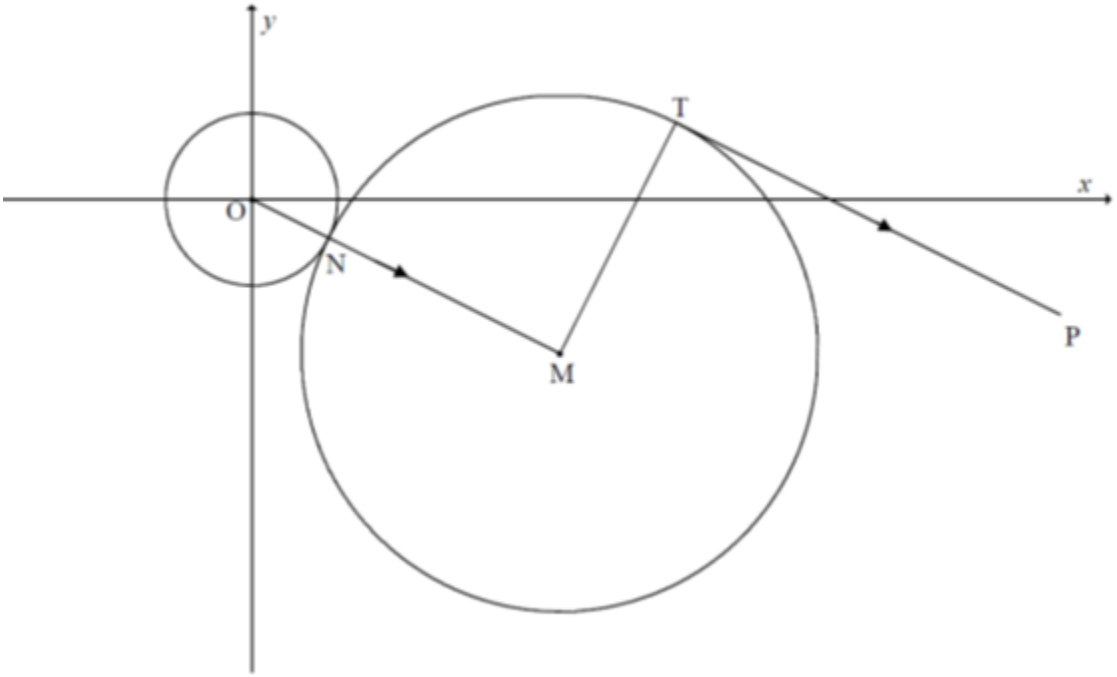
14.5 Write down the value of α, the inclination of line KM. (1)

14.6 If points R$(1; k)$, J and L are collinear, calculate the value of k. (4)

[21]

Activity 15

In the diagram below, the equation of the circle with centre M is $(x-8)^2+(y+4)^2=45$. PT is a tangent to this circle at T and PT is parallel to OM. Another circle, having centre O, touches the circle having centre M at N.



15.1 Write down the coordinates of M. (1)

15.2 Calculate the length of OM. Leave your answer in simplest surd form. (2)

15.3 Calculate the length of ON. Leave your answer in simplest surd form. (3)

15.4 Calculate the size of $O\widehat{M}T$. (2)

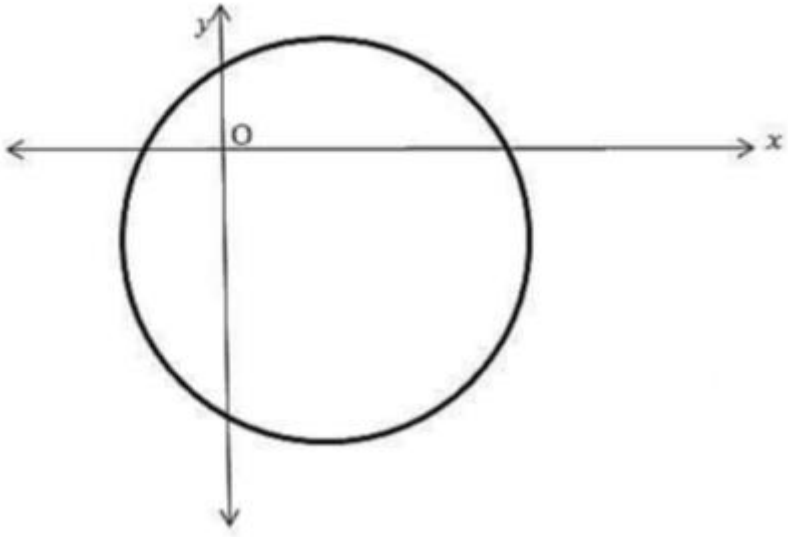
15.5 Determine the equation of MT in the form $y=mx+c$. (5)

15.6 Calculate the coordinates of T. (6)

[19]

Activity 16

The length of the radius of the circle with equation $x^2 - 2x + y^2 + 4y = a$ is 5 units.



16.1 Show by means of calculations that $a = 20$ units. (4)

16.2 Write down the coordinate of the centre M of the circle. (1)

16.3 A$(x; y)$, with $y > 0$, is one of the ponts of intersection of the circle and the straight line $x = 4$. Determine the values of x and y. (4)

16.4 Determine the equation of the tangent to the circle at point A. (4)

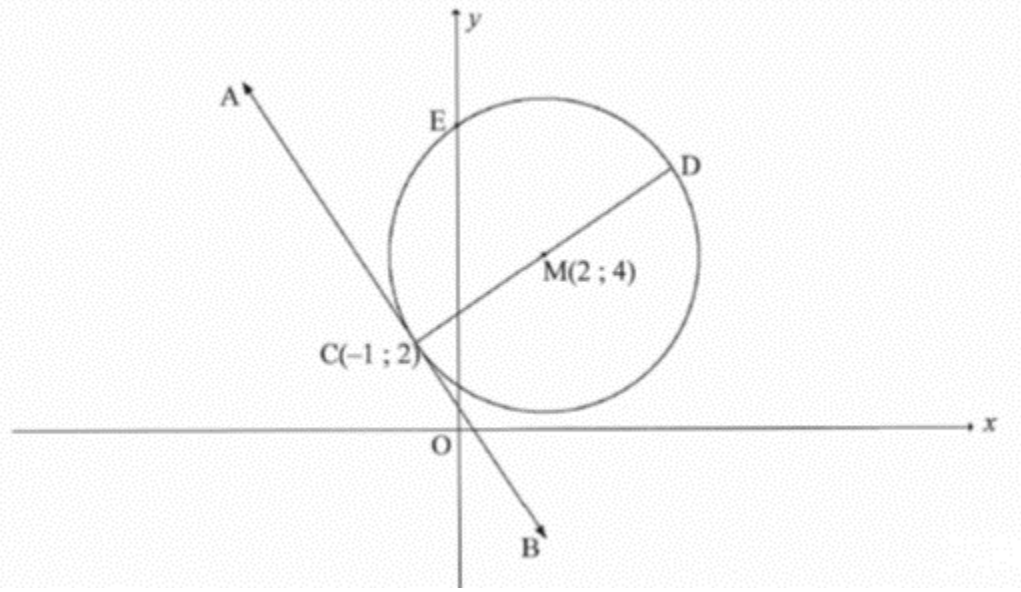
16.5 Determine whether point T$(-1; -2)$ lies inside or outside the circle. (3)

16.6 If the circle is translated 3 units to the left and l unit up, determine the equation of the new circle. (3)

[19]

Activity 17

17.1 In the diagram below, the circle centred at $M(2;4)$ passes through $C(-1;2)$ cuts the y-axis at E. the diameter CMD is drawn and ACB is a tangent to the circle.



17.1.1 Determine the equation of the circle in the form $(x-a)^2+(y-b)^2=r^2$. (3)

17.1.2 Write down the coordinates of D. (2)

17.1.3 Determine the equation of AB in the form $y=mx+c$. (5)

17.1.4 Calculate the coordinates of E. (4)

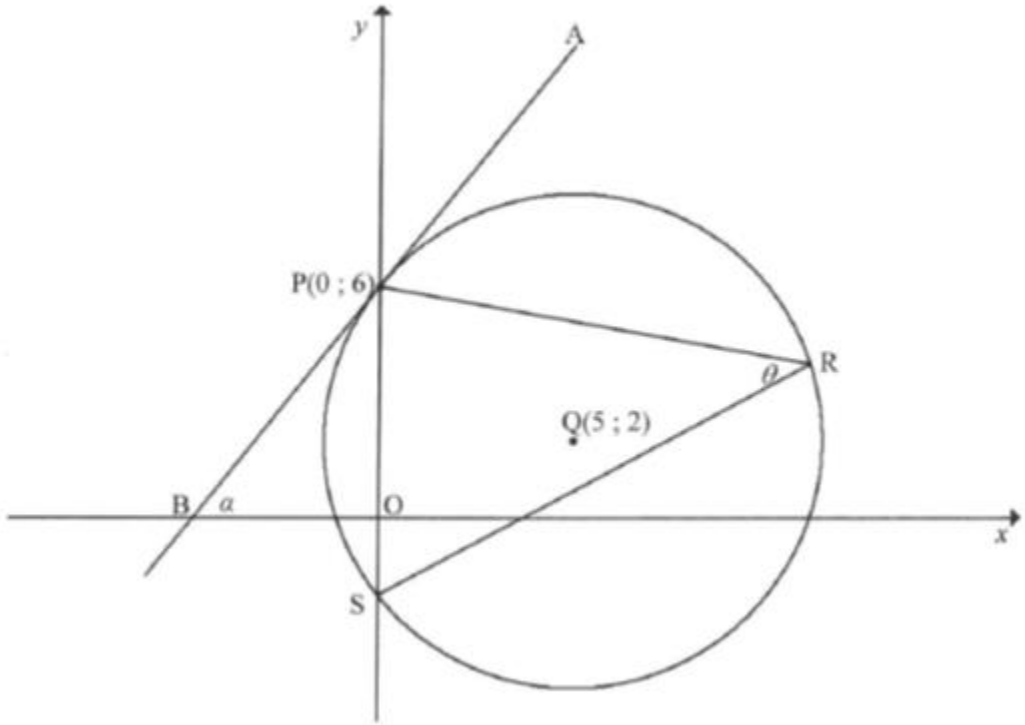
17.1.5 Show that EM is parallel to AB. (2)

17.2 Determine whether or not the circles having equations $(x+2)^2+(y-4)^2=25$ and $(x-5)^2+(y+1)^2=9$ will intersect. Show ALL calculations. (6)

[22]

Activity 18

In the diagram below, $Q(5;2)$ is the centre of a circle that intersects the y-axis a $P(0;6)$ and S. the tangent APB at P intersects the x-axis at B and makes the angle α with the positive x-axis. R is a point on the circle and $P\widehat{R}S = \theta$.



18.1 Determine the equation of the circle in the form $(x-a)^2+(y-b)^2=r^2$. (3)

18.2 Calculate the coordinates of S. (3)

18.3 Determine the equation of the tangent APB in the form $y=mx+c$. (4)

18.4 Calculate the size of α. (2)

18.5 Calculate, with reasons, the size of θ. (4)

18.6 Calculate the area of ΔPQS. (4)

[20]

Activity 1

1.1	$r^2 = (0-12)^2 + (0-5)^2$ $r^2 = 169$ $x^2 + y^2 = 169$
1.2	$y - y_1 = m(x - x_1)$ $y - 5 = \frac{12}{5}(x - 12)$ $y = \frac{12}{5}x$
1.3	$P(-12; -5)$
1.4	$PQ \perp QR$ $m_{QR} \times m_{PQ} = -1$ $m_{QR} \times \frac{5}{12} = -1$ $m_{QR} = -\frac{12}{5}$
1.5	$y - y_1 = m(x - x_1)$ $y - 5 = -\frac{12}{5}(x - 12)$ $y = -\frac{12}{5}x + \frac{169}{5}$
1.6	$y = -\frac{12}{5}x + \frac{169}{5}$ $-1 = -\frac{12}{5}(t) + \frac{169}{5}$ $t = \frac{29}{2}$
1.7	$(x-12)^2 + (y-5)^2 = 169$

Activity 2

2.1	$M\left(\frac{x_1+x_2}{2};\frac{y_1+y_2}{2}\right)$ $M\left(\frac{-8+0}{2};\frac{1+5}{2}\right)$ $M(-4;3)$
2.2	$m_{BC}=\frac{y_2-y_1}{x_2-x_1}$ $m_{BC}=\frac{5+1}{0-3}$ $m_{BC}=-2$ $y-y_1=m(x-x_1)$ $y-5=-2(x-0)$ $y=-2x+5$
2.3	$d=\sqrt{(y_2-y_1)^2+(y_2-y_1)^2}$ $AM=\sqrt{(-4-0)^2+(3-5)^2}$ $AM=2\sqrt{5}$ units
2.4	$(x+4)^2+(y-3)^2=20$ $x^2+8x+16+y^2-6y+9=20$ $x^2+y^2+8x-6y-11=0$
2.5	Kite, one pair of opposite angles are 90° each. Diagonals are perpendicular to each other, $CD=AK$

Activity 3

3.1	$x^2 + y^2 + 8x - 6y = -5$ $x^2 + 8x + 16 + y^2 - 6y + 9 = -5 + 16 + 9$ $(x + 4)^2 + (y - 3)^2 = 20$ $\therefore M(-4; 3)$
3.2	$(0 + 4)^2 + (y - 3)^2 = 20$ $16 + y^2 - 6y + 9 - 20 = 0$ $y^2 - 6y + 5 = 0$ $(y - 1)(y - 5) = 0$ $\therefore Q(0,1)$
3.3	$m_{UE} \times m_{DQ} = -1$ $m_{UE} \times \frac{6 - 1}{-10 - 0} = -1$ $m_{UE} = 2$ $y - y_1 = m(x - x_1)$ $y - 1 = 2(x - 0)$ $y = 2x + 1$
3.4	$y = 6$
3.5	$6 = 2x + 1$ $x = \frac{5}{2}$ $D\left(\frac{5}{2}; 6\right)$
3.6	$D\hat{Q}U = 90°$ $\quad (tangent \perp radius)$ $m_{AD} \times m_{AU}$ $= \frac{11 - 6}{0 - 10} \times \frac{11 - 6}{0 - \left(\frac{5}{2}\right)}$ $= -1$

	$m_{AD} \times m_{AU} = -1$ $\therefore D\hat{A}U = 90°$ $D\hat{A}U + D\hat{Q}U = 180°$ Converse of opposite angles of a cyclic quadrilateral.

Activity 4

4.1	$x^2 + 8x + 16 + y^2 + 4y + 4 = 38 + 16 + 4$ $(x + 4)^2 + (y + 2)^2 = 58$ $C(-4; -2)$ $r = \sqrt{58}$
4.2	$C_1(-4; -2)$ $C_2(4; 6)$ $C_1C_2 = \sqrt{(-4 - 4)^2 + (-2 - 6)^2}$ $C_1C_2 = 8\sqrt{2}$ units $C_1C_2 = 11{,}31$ units
4.3	$R = r_2 - r_1$ $R = \sqrt{58} - \sqrt{26}$ $R = 2{,}52$ units $C_1C_2 > R$ $\therefore$ The two circles intersect.
4.4	$x^2 + y^2 + 8x + 4y - 38 = 0$ …….(1) $x^2 + y^2 - 8x - 12y - 46 = 0$ ……...(2) $(1) - (2)$: $16x + 16y = 8$

	$\therefore y = -x + \frac{1}{2}$

Activity 5

5.1	$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$ $M\left(\frac{0 - 2}{2}; \frac{8 - 6}{2}\right)$ $M(-1; 1)$
5.2	$y = 7(-8) + 58$ $y = 2$
5.3	They intersect at $(-8; 2)$ $m_{AM} = \frac{y_2 - y_1}{x_2 - x_1}$ $m_{AM} = \frac{2 - 1}{-8 + 1}$ $m_{AM} = -\frac{1}{7}$ $l \perp AM$ $\therefore y = 7x + 58$ is a tangent to the circle.
5.4	$d = \sqrt{(y_2 - y_1)^2 + (y_2 - y_1)^2}$ $AD = \sqrt{(-8 - 0)^2 + (2 - 8)^2}$ $AD = 10$ units $AB = \sqrt{(-8 + 2)^2 + (2 + 6)^2}$ $AB = 10$ units
5.5	$m_{AD} \times m_{AU}$

	$= \frac{8-2}{0+8} \times \frac{2+6}{-8+2}$ $= -1$ $AD \perp AB$ $\therefore D\hat{A}B = 90°$
5.6	$\theta = 45°$

Activity 6

6.1.1	$(2)^2 + y^2 - 8(2) + 6y - 15 = 0$ $y^2 + 6y - 27 = 0$ $(y-3)(y+9) = 0$ $y = 3$ or $y = -9$ $\therefore (2; -9)$ lies on the circle
6.1.2	$x^2 - 8x + 16 + y^2 + 6y + 9 = 15 + 16 + 9$ $(x-4)^2 + (y+3)^2 = 40$ $C(4; -3)$ $m_{rad} = \frac{-9+3}{2+3}$ $m_{rad} = -\frac{6}{5}$ $m_{tan} = \frac{5}{6}$ $y - y_1 = m(x - x_1)$ $y + 9 = \frac{5}{6}(x-2)$ $y = \frac{5}{6}x + \frac{22}{3}$

Activity 7

7.1	$m_{tan} = -1$ $\therefore m_{rad} = 1$ $y - y_1 = m(x - x_1)$ $y - 4 = 1(x + 4)$ $y = x + 8$
7.2	$-x + 2 = x + 8$ $-2x = 6$ $x = -3$ $y = -3 + 8$ $y = 5$ $\therefore L(-3; 5)$
7.3	$r^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$ $r^2 = (-3 + 4)^2 + (5 - 4)^2$ $r^2 = 2$ $(x + 4)^2 + (y - 4)^2 = 2$
7.4	$\frac{x_N - 3}{2} = -4$ $x_N = -5$ $\frac{y_N + 5}{2} = 4$ $y_N = 3$ $\therefore N(-5; 3)$
7.5	$y = -x - 2$

Activity 8

8.1	$r^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$ $r^2 = (1 + 2)^2 + (-2 - 1)^2$ $r^2 = 3\sqrt{2}$ $M(-2; 1)$
8.2	$m_{rad} \times m_{tan} = -1$ $\frac{1 + 2}{-2 - 1} \times m_{tan} = -1$ $m_{tan} = 1$ $y - y_1 = m(x - x_1)$ $y + 2 = 1(x - 1)$ $y = x - 3$
8.3	Equation of MS: $y - 1 = -1(x + 2)$ $y = -x - 1$ $b = -a - 1$ ……..(1) $MP = 9\sqrt{2}$ $MP^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$ $\left(9\sqrt{2}\right)^2 = (-2 - a)^2 + [1 - (-a - 1)]^2$ $162 = a^2 + 4a + 4 + a^2 + 4a + 4$ $a^2 + 4a - 77 = 0$ $(a - 7)(a + 11) = 0$ $a = 7$ or $a = -11$ $\therefore a = 7$

	$b = -(7) - 1$ $b = -8$ $\therefore P(7; -8)$

Activity 9

9.1	$Tangent \perp Radius$
9.2	$L(3; 0)$
9.3	$(x-3)^2 + (y-2)^2 = 4$
9.4	$0 = \frac{4}{3}x + \frac{4}{3}$ $x = -1$ $K(-1; 0)$ $\therefore KL = 4$ units
9.5	$y - y_1 = m(x - x_1)$ $y - 2 = -\frac{3}{4}(x - 2)$ $y = -\frac{3}{4}x + \frac{17}{4}$
9.6	$-\frac{3}{4}x + \frac{17}{4} = \frac{4}{3}x + \frac{4}{4}$ $-\frac{25}{12}x = -\frac{35}{12}$ $x = \frac{7}{5}$ $y = -\frac{3}{4}\left(\frac{7}{5}\right) + \frac{17}{4}$ $y = \frac{16}{5}$

	$\therefore A\left(\frac{7}{5};\frac{16}{5}\right)$
9.7	$KA = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $KA = \sqrt{\left(\frac{7}{5} + 1\right)^2 + \left(\frac{16}{5} - 0\right)^2}$ $KA = 4$ units
9.8	$AN = NL$ (Adjacent sides are equal) $AK = KL$ (Adjacent sides are equal) $AK \perp AN$ and $KL \perp NL$ (One pair of opposite angles at 90° each)
9.9	ΔKAB is a right-angle triangle which is isosceles, because $AB = AK$ $\therefore \widehat{ABK} = 45°$
9.10	$N'(3;-2)$

Activity 10

10.1.1	$x^2 + y^2 = r^2$ $(2)^2 + (-1)^2 = r^2$ $r = \sqrt{5}$ units
10.1.2	$ED = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $2\sqrt{5} = \sqrt{(a-2)^2 + (-3-0)^2}$ $20 = a^2 - 4a + 4 + 4$ $a^2 - 4a - 12 = 0$ $(a-6)(a+2) = 0$ $a = 6$ or $a = -2$

	$\therefore a = 6$
10.1.3	$(x-6)^2 + (y+3)^2 = 20$
10.1.4	$m_{rad} \times m_{tan} = -1$ $\frac{-1+3}{2-6} \times m_{tan} = -1$ $m_{tan} = 2$ $y - y_1 = m(x - x_1)$ $y + 1 = 2(x-2)$ $y = 2x - 5$
10.2.1	$x^2 - 4x + y^2 + 5y = k$ $(x-2)^2 + \left(y + \frac{5}{2}\right)^2 = k + 4 + \frac{25}{4}$ $\therefore C\left(2; -\frac{5}{2}\right)$
10.2.2	$k + 4 + \frac{25}{2} = r^2$ $k + 4 + \frac{25}{4} = (12)^2$ $k = \frac{535}{4}$

Activity 11

11.1.1	$(x+1)^2+(y+3)^2=8$ $M(-1;3)$
11.1.2	$r_1=2\sqrt{2}$ units
11.2	$y=x-2$ ……..(1) $(x+1)^2+(y+3)^2=8$ ……..(2) Substitute y into (2): $(x+1)^2+(x-2+3)^2=8$ $x^2+2x+1+x^2+2x+1=8$ $2x^2+4x-6=0$ $x^2+2x-3=0$ $(x+3)(x-1)=0$ $x=-3$ or $x=1$ $\therefore x=-3$ $y=-3-2$ $y=-5$ $D(-3;-5)$
11.3	$MB^2=DM^2+BD^2$ $MB^2=\left(2\sqrt{2}\right)^2+\left(4\sqrt{2}\right)^2$ $MB=2\sqrt{10}$ units
11.4	$(x+1)^2+(y+3)^2=40$
11.5	$MF=\sqrt{\left(-1-2\sqrt{5}\right)^2+(-3-0)^2}$ $MF=6{,}24$

	$MB = 5{,}66$ $MF > MB$ $\therefore$ F lies outside C_2

Activity 12

12.1	$(-1-3)^2 + (4-1)^2 = r^2$ $25 = r^2$ $(x+1)^2 + (y-4)^2 = 25$
12.2	Angle subtended by the diameter OR Angle in the semi-circle
12.3	$(x+1)^2 + (0-4)^2 = 25$ $(x+1)^2 = 9$ $x + 1 = \pm 3$ $x + 1 = 3$ or $x + 1 = -3$ $x = 2$ or $x = -4$ $\therefore Q(-4;0)$
12.4	$m_{MN} = \frac{4-1}{-1-3}$ $m_{MN} = -\frac{3}{4}$
12.5	$\tan\beta = m_{MQ}$ $\beta = \tan^{-1}\left(\frac{1-0}{3+4}\right)$ $\beta = 8{,}13°$ $\theta = \tan^{-1}\left(-\frac{3}{4}\right) + 180°$ $\theta = 143{,}13°$

	$\alpha + (90° + \beta) = 180°$ (Exterior angle of a triangle) $\alpha = 45°$
12.6	$m_{tan} = \frac{4}{3}$ $y - y_1 = m(x - x_1)$ $y - 1 = \frac{4}{3}(x - 3)$ $y = \frac{4}{3}x - 3$

Activity 13

13.1	$Q(x; 2)$ $3x + 4y + 7 = 0$ $3x + 4(2) + 7 = 0$ $x = -3$ $Q(-3; 2)$ $r = QP$ $r = 3$ units $\therefore (x + 3)^2 + (y - 2)^2 = 9$
13.2	$WZ = 2 \times QP$ $\therefore WZ = 6$ units

Activity 14

14.1	$m_{JK} = m_{ML}$ $\frac{1+4}{-2-p} = \frac{2-0}{3-5}$ $\frac{5}{-2-p} = -1$ $-1(-2-p) = 5$ $\therefore p = 3$
14.2	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $JK = \sqrt{(3+2)^2 + (-4-1)^2}$ $JK = 5\sqrt{2}$ units $ML = \sqrt{(5-3)^2 + (0-2)^2}$ $ML = 2\sqrt{2}$ units $JK : ML = 5 : 2$
14.3	For the equation of QL: $m_{QL} = m_{JM}$ $\frac{y-0}{x-5} = \frac{2-1}{3+2}$ $5y = x - 5$ $x = -5y + 5$ ……(1) For the equation of JK: $m_{JK} = \frac{1+4}{-2-3}$ $m_{JK} = -1$ $y - 1 = -1(x+2)$ $y = -x - 1$ ……(1) $x = 5(-x-1) + 5$

	$x = 0$ $y = -(0) - 1$ $y = -1$ $\therefore Q(0; -1)$
14.4	$x = 3$
14.5	$\alpha = 90°$
14.6	$m_{JR} = m_{RL}$ $\frac{k-1}{1+2} = \frac{0-1}{5+2}$ $\frac{k-1}{3} = -\frac{1}{7}$ $\therefore k = \frac{4}{7}$

Activity 15

15.1	$M(8; -4)$
15.2	$OM = \sqrt{(0-8)^2 + (0+4)^2}$ $OM = 4\sqrt{5}$ units
15.3	$ON = 0M - MN$ $ON = 4\sqrt{5} - 3\sqrt{5}$ $ON = \sqrt{5}$ units
15.4	$M\hat{T}P = 90°$ $(Tangent \perp Radius)$ $O\hat{M}T = M\hat{T}P$ (Alternate angles, OM||TP) $O\hat{M}T = 90°$

15.5	$MT \perp OM$ $m_{OM} = \frac{-4-0}{8-0}$ $m_{OM} = -\frac{1}{2}$ $m_{MT} = 2$ $y - y_1 = m(x - x_1)$ $y + 4 = 2(x - 8)$ $y = 2x - 20$
15.6	$y = 2x - 20$ ……(1) $(x-8)^2 + (y+4)^2 = 45$ ……(2) Substitute (1) into (2): $(x-8)^2 + (2x - 20 + 4)^2 = 45$ $(x-8)^2 + (2x-16)^2 = 45$ $x^2 - 80x + 64 + 4x^2 - 64x + 256 = 45$ $5x^2 - 80x + 275 = 0$ $x^2 - 16x + 55 = 0$ $(x-11)(x-5) = 0$ $x = 11$ or $x = 5$ $\therefore T(11;2)$

Activity 16

16.1	$x^2 - 2x + y^2 + 4y = a$ $(x-1)^2 + (y+2)^2 = a + 1 + 4$ $(x-1)^2 + (y+2)^2 = a + 5$ $r^2 = a + 5$ $5^2 = a + 5$ $a = 20$
16.2	$M(1;-2)$
16.3	$(x-1)^2 + (y+2)^2 = 25$ $(4-1)^2 + (y+2)^2 = 25$ $9 + y^2 + 4y + 4 = 25$ $y^2 + 4y - 12 = 0$ $(y+6)(y-2) = 0$ $y = -6$ or $y = 2$ $\therefore A(4;2)$
16.4	$m_{rad} = \frac{-2-2}{1-4}$ $m_{rad} = \frac{4}{3}$ $m_{tan} = -\frac{3}{4}$ $y - y_1 = m(x - x_1)$ $y + 4 = -\frac{3}{4}(x-8)$ $y = -\frac{3}{4}x + 2$

16.5	$r = 5$ units $MT = \sqrt{(1+1)^2 + (-2+2)^2}$ $MT = 2$ units $MT < r$ ∴ T lies inside the circle
16.6	$(x+2)^2 + (y+1)^2 = 25$

Activity 17

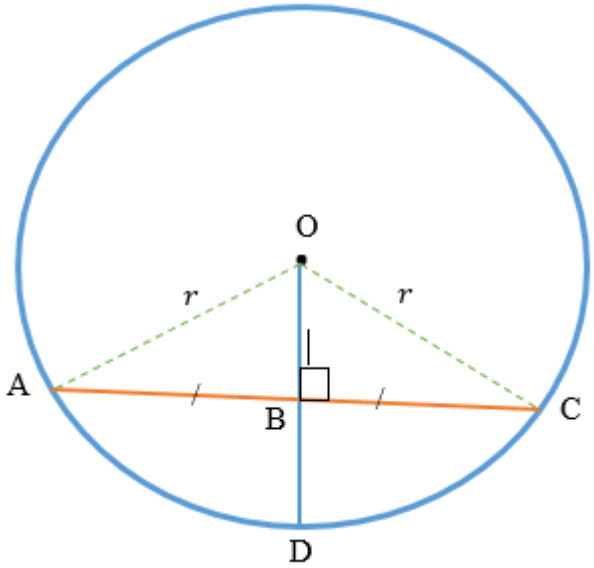
17.1.1	$(1+2)^2 + (4-2)^2 = r^2$ $13 = r^2$ $(x-2)^2 + (y-4)^2 = 13$
17.1.2	$D(5;6)$
17.1.3	$m_{rad} = \frac{4-2}{2+1}$ $m_{rad} = \frac{2}{3}$ $m_{tan} = -\frac{3}{2}$ $y - y_1 = m(x - x_1)$ $y - 2 = -\frac{3}{2}(x+1)$ $y = -\frac{3}{2}x + \frac{1}{2}$
17.1.4	$(x-2)^2 + (y-4)^2 = 13$ $(0-2)^2 + (y-4)^2 = 13$ $4 + y^2 - 8y + 16 = 13$

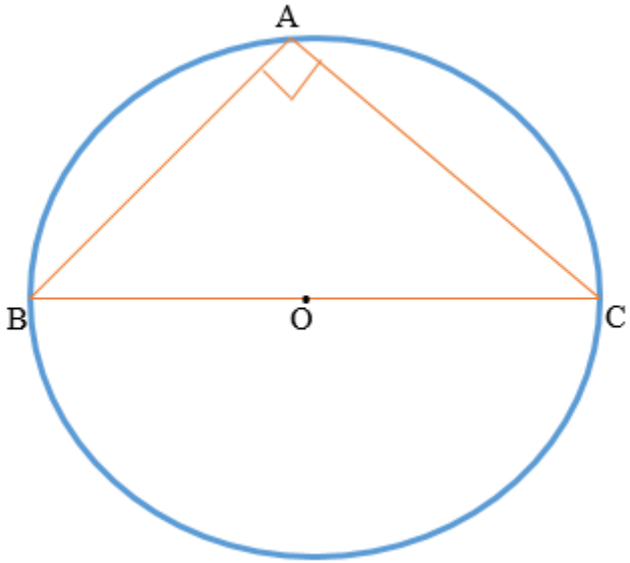
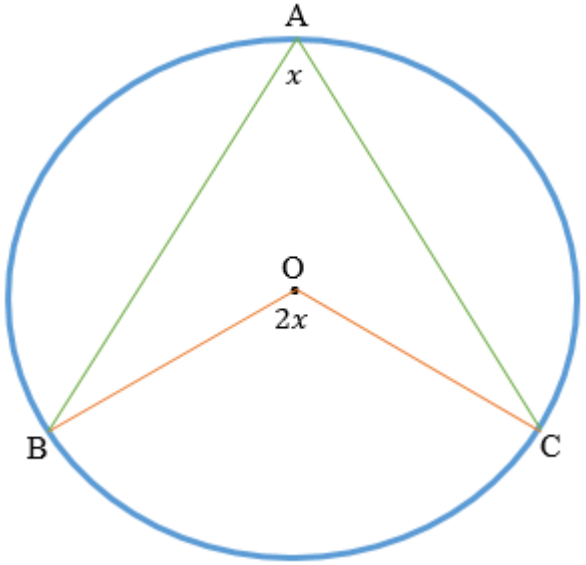
	$y^2 - 8y + 7 = 0$ $(y - 7)(y - 1) = 0$ $y = 7$ or $y = 1$ $\therefore E(0; 7)$
17.1.5	$m_{EM} = \frac{7 - 4}{0 - 2}$ $m_{EM} = -\frac{3}{2}$ $m_{EM} = m_{AB}$ $\therefore EM \| \| AB$
17.2	$C_1(-2; 4)$ and $r_1 = 5$ units $C_2(5; -1)$ and $r_2 = 3$ units $C_1C_2 = \sqrt{(-2 - 5)^2 + (4 + 1)^2}$ $C_1C_2 = 9$ units $r_1 + r_2 = 8$ units $C_1C_2 > r_1 + r_2$ $\therefore$ The two circles do not intersect.

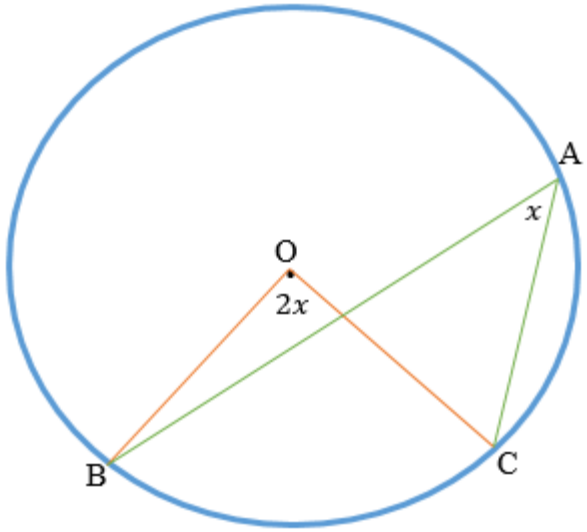
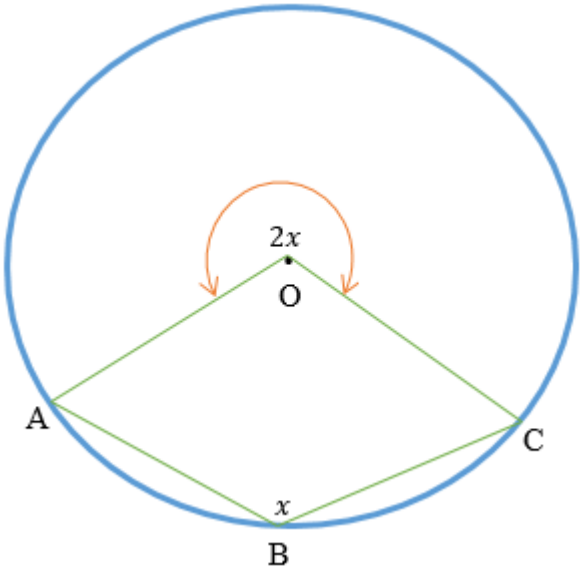
Activity 18

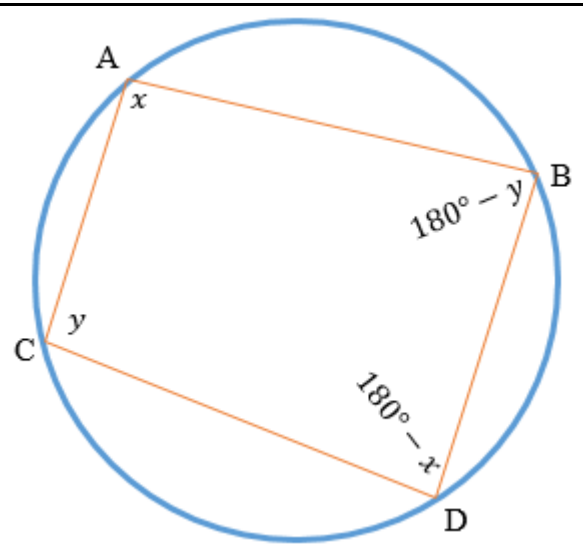
18.1	$(5-0)^2+(2-6)^2=r^2$ $41=r^2$ $(x-5)^2+(y-2)^2=41$
18.2	$(-5)^2+(y-2)^2=41$ $25+y^2-8y+4=41$ $y^2-4y-12=0$ $(y-6)(y+2)=0$ $y=6$ or $y=-2$ $\therefore S(0;-2)$
18.3	$m_{rad}=\frac{6-2}{0-5}$ $m_{rad}=-\frac{4}{5}$ $m_{tan}=\frac{5}{4}$ $y-y_1=m(x-x_1)$ $y-6=\frac{5}{4}(x-0)$ $y=\frac{5}{4}x+6$
18.4	$\tan\alpha=m_{AB}$ $\alpha=\tan^{-1}\left(\frac{5}{4}\right)$ $\therefore \alpha=51{,}34°$

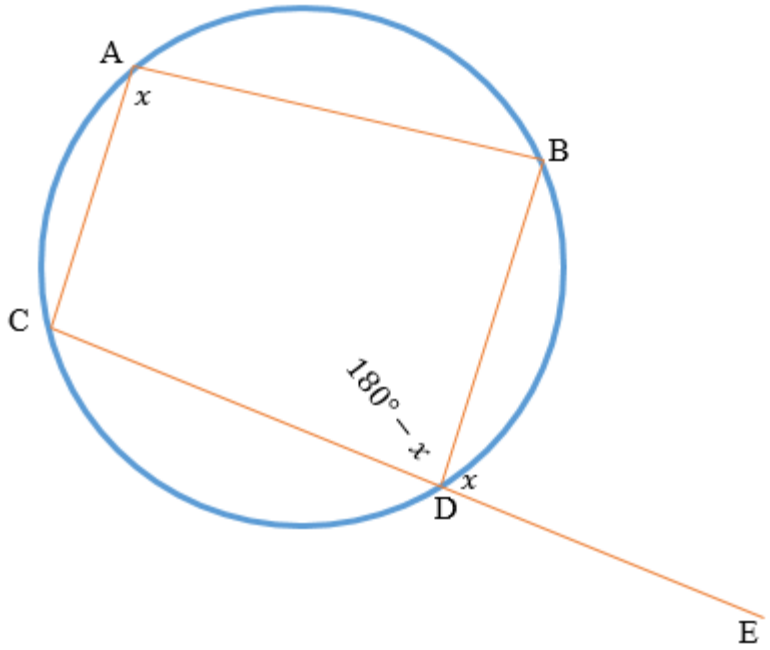
18.5	In ΔPQS $PQ = \sqrt{41}$ $QS = \sqrt{41}$ $PS = 8$ $\cos \hat{Q} = \frac{PQ^2 + QS^2 - PS^2}{2(PQ)(QS)}$ $\cos \hat{Q} = \frac{\left(\sqrt{41}\right)^2 + \left(\sqrt{41}\right)^2 - (8)^2}{2(\sqrt{41})(\sqrt{41})}$ $\hat{Q} = 77{,}32°$ $\hat{Q} = 2\theta$ ($\angle$ at centre $= 2 \times \angle$ at circumference) $77{,}32° = 2\theta$ $\theta = 38{,}66°$
18.6	$Area_{\Delta PQS} = \frac{1}{2}bh$ $Area_{\Delta PQS} = \frac{1}{2}(8)(5)$ $Area_{\Delta PQS} = 20 \text{ units}^2$

CONCEPT	HOW TO LEARN IT?	RELEVANT FORMULAE AND KEYWORDS
Theorem 1: Line segment through centre, mid-point and converse. **Theorem 2:** Perpendicular bisector of a chord.	Learn and use Congruency; 90°; H; S or A; A; S Lookout for the midpoint, then conclude 90° at the midpoint.	
	This theorem brings a right-angle triangle in the picture, so Pythagoras is important and the midpoint (as in Analytical Geometry).	Look out for the centre or midpoint on a chord as well as mid-points of two sides of a triangle. Look out for the centre
	Congruency to prove the theorem (Theorem 1). The use of Pythagoras as this theorem brings along 90° angle and a right-angled triangle. Ensure that two sides of a triangle are bisected by the same line. First Midpoint of a line, then 90° will follow.	

	You need to ensure that there is a centre and 90°, then the line will pass through the centre.	
	If $\hat{A} = \hat{B}$ and $\hat{B} = \hat{C}$, then $\hat{A} = \hat{C}$ If $\hat{A} + \hat{B} = \hat{C}$ and $\hat{A} + \hat{D} = \hat{C}$, then $\hat{B} = \hat{C}$. Applications of these rules without numerical values, that is given variables such as x or y. Expressing one variable in terms of the other information, that is changing the subject of the formula of the equation. Substitution of equal quantities.	
Theorem 3: Angle at centre is twice angle at circumference. Diameter subtends right angle at circumference.	Learn exterior angle of a triangle and properties of an Isosceles triangle. The proof of this theorem is for examination purpose. (That is, it is a possible exam question). 	

	Use the previous theorem or lookout for the centre, diameter and the angle subtended by the line that passes through the centre (diameter).	 

	Arc: The angle at the centre and the one on the circle must be on the same arc or chord. This is a special case of the above theorem; it also brings about a 90° angle into the picture a right-angled triangle.	Centre Centre
	Always lookout for an arc that supports the angle at the centre, then look for the angle at the circle that is also supported by the arc. (Think about subtend or support, as a shouting mouth, shouting at the arc or chord).	
Opposite angles of a cyclic Quadrilateral Exterior angle of a cyclic quadrilateral. Proving that a quadrilateral is cyclic.	The proof of this theorem is for examination purpose. For the proof use the theorem about the angle at the centre and a revolution. Look out for the straight line coming from one of the vertices of the cyclic quad.	

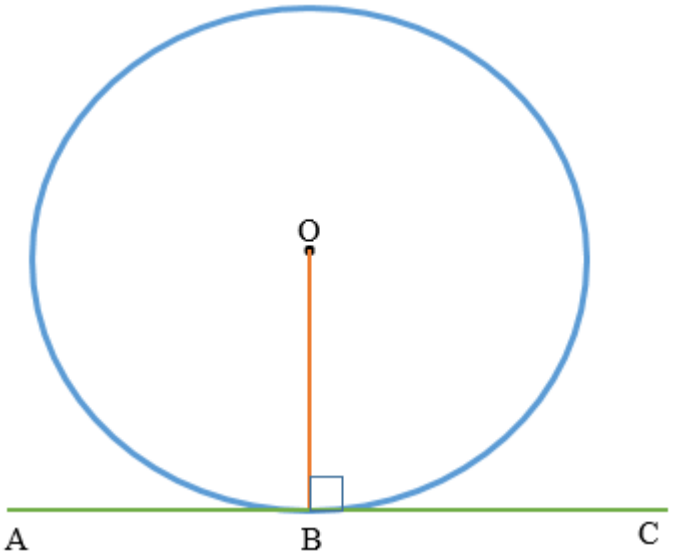


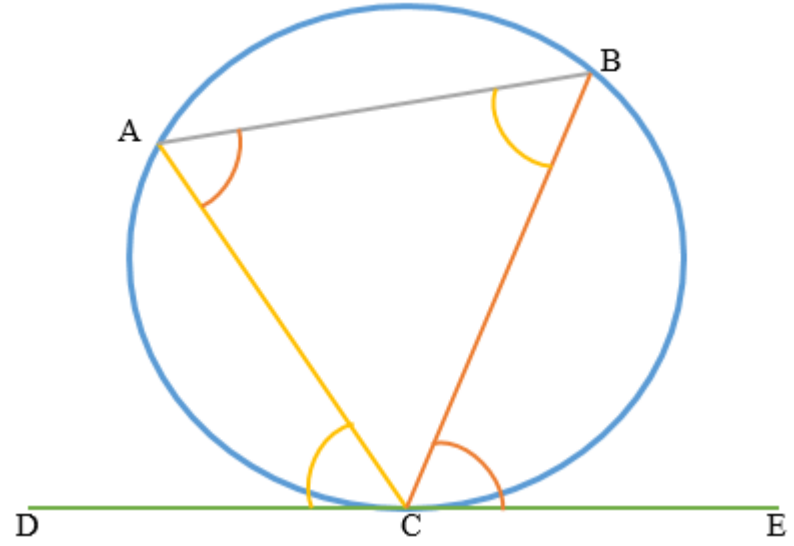
This theorem brings = 180° into the picture.
There are three options:

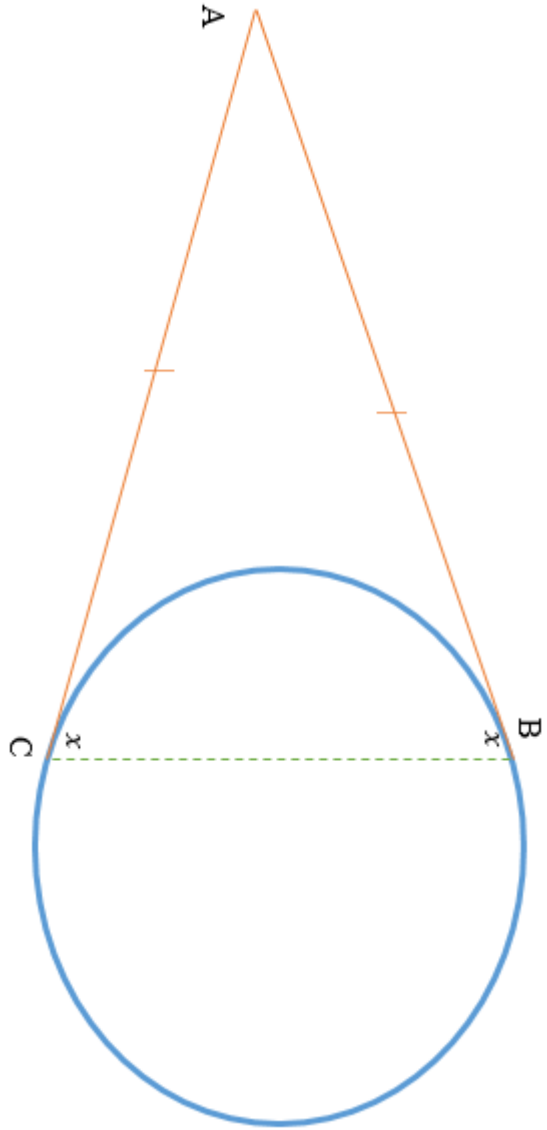
Lookout to see if you can have opposite angles of a
quad = 180°, then that quad will be cyclic.

Lookout to see if you can have ext. angle of a quad = interior opposite angle, then the quad will be cyclic.

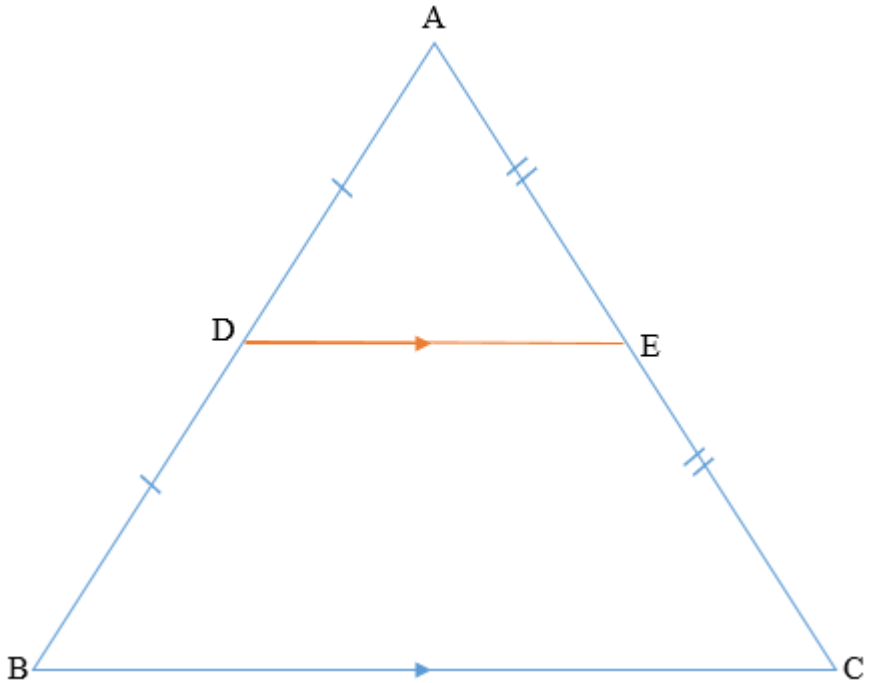
Lookout to see if you can have two angles that are subtended by a line, then the four vertices will be a cyclic quad.

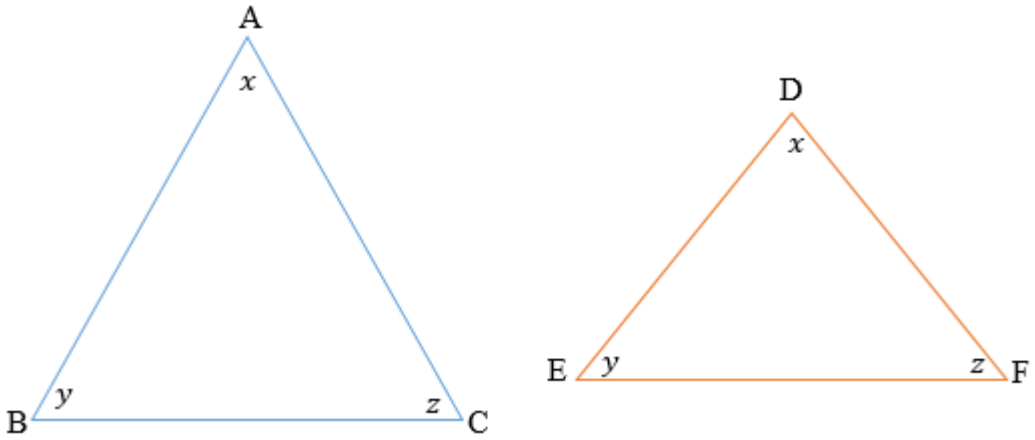
	Two properties of the cyclic quad.	
Tan-chord theorem Tangents from same point Tangent perpendicular to radius	Proof for examination purposed. To prove: use the diameter (angle at the centre), Tangent Radius Theorem (Tan – Radius), sum of angles of a triangle and second logical reason of addition.	

		 A B D C E


A
B
C
x
x

	Divide the circle at the point of contact (between the chord and the tangent) to be able to identify the alternate segment, then focus on the chord and look for any angle that is supported by the chord in the other segment.	Tangent (see solving riders).
	Three properties of the tangent to a circle.	
Solving riders	Lookout for the key words connected to theorems in the explicitly given information, for instance, centre (connected theorems, include theorem 1, 2, angle at the centre together(diameter). Tangent (connected theorems include, (tan-chord, tan-radius and tangents drawn from the same point) and Cyclic Quad. (Connected theorems, include opposite angles are supplementary and ext. angle of a cyclic quad = to the opposite interior angle). Examine the implicitly given information from the diagram, then make conclusion on what you get from the diagram. Create a short list of statements and reasons, see if you cannot make logical conclusion from the list using important logical reasoning. Lookout for the angles on the same segment in the diagram.	

	Lookout for angles on equal chords (the angles are equal). Then answer the questions.	
	See above Solving riders.	
	The more you practice, the better you become.	
Theorem 1: Line drawn parallel to one side of triangle. The midpoint theorem	For examination purpose. Remember that a triangle can have its height inside the triangle or outside the triangle. The area of triangle between parallel lines and share a base have the equal area. After the constructions rotate the two sides that are not parallel to any of the lines so that it is horizontal, you will see the heights that are being shared by different triangles. Lookout for the midpoint of two sides of the triangle.	
	Properties (the conclusion of the theorem, the part that follows the word **then** in the theorem) of the theorem.	

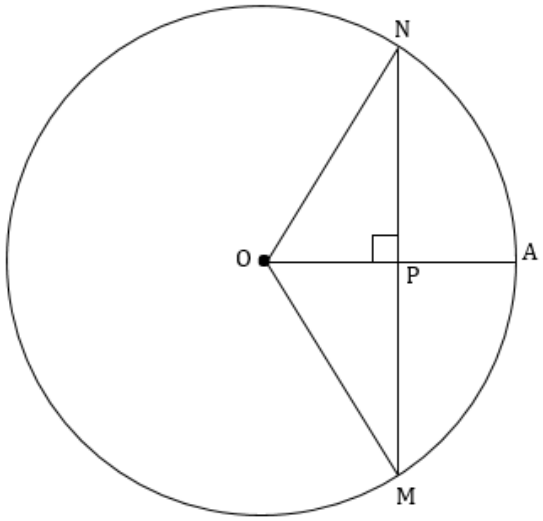
Theorem 2: Equiangular triangles are similar.	Examination purpose. Use congruency S: A: S to create a line parallel to one side of a triangle, so that you can form proportions using one triangle. Application: Remember that this theorem requires that you use more than one triangle and the order in naming the triangles is also important as well. **Otherwise, lookout for corresponding sides to prove similarity.**  $\Delta ABC \vert\vert\vert \Delta DEF$	
	Do plenty of examples.	
Theorem of Pythagoras	Proof is not required for examination purpose. Otherwise, first write down the sum of the squares of the two side = the square of the hypotenuse. Then look out for either equal sides from the diagram or the sum of two sides making one side.	
	Do plenty of examples.	
Proportion	Work with one triangle	

Similarity	Work with two triangles. Remember after similarity (equiangular) you also get proportion, in this case use two triangles. If asked to prove proportion look for the allocated marks, if the marks are at least 5, then start by proving similarity using (equiangular) by rearranging the proportion as follows, the first side on the right becomes the denominator on the left and the second side on the left becomes the denominator on the right, in that way you the numerators come from the same triangle and the denominators come from the second triangle. E.g. if asked to prove: $AB \times PQ = PR \times BC$ Then rearranging becomes: $\frac{AB}{PR} = \frac{BC}{PQ}$ so that the triangles are triangle ABC the numerators and triangle PQR from the denominators.	

TUTORIAL 1

Activity 1

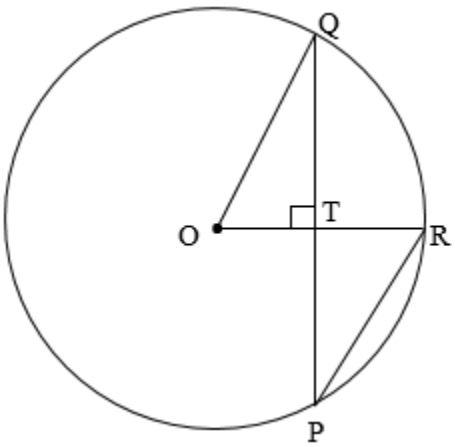
In the diagram, O is the centre of the circle NAM and OPA ⊥ MPN. MN = 48 units and OP = 7 units.



Calculate, with reason, the length of PA. **[5]**

Activity 2

In the diagram below, PQ is the chord of circle O. OR⊥PQ and OR intersects PQ at T. If the radius of the circle is 13 cm and PT = 12 cm.



Calculate the length of:

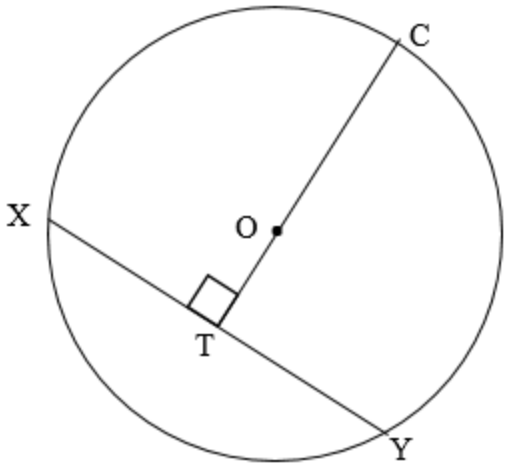
2.1 PQ (2)

2.2 PR (4)

Activity 3

In the diagram drawn below, O is the centre of the circle XCY. COT $\perp$ XY.

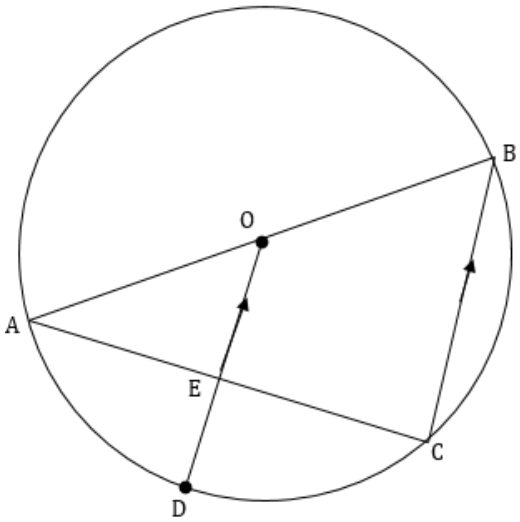
OC $= r$ and XY $= \frac{3}{2}r$.



Prove, stating reasons, that $\text{CT} = \frac{4+\sqrt{7}}{4}r$. **[6]**

Activity 4

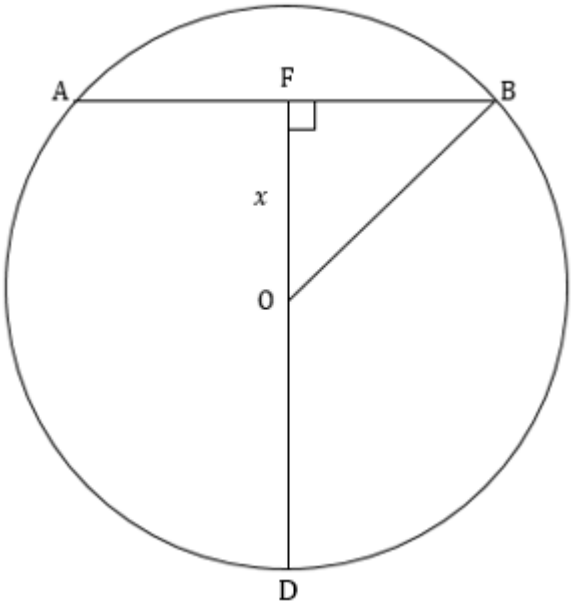
AB is a diameter of the circle ABCD. OD is drawn parallel to BC and meets AC at E.



If the radius is 10 cm and AC $= 16$ cm, calculate the length of ED. **[5]**

Activity 5

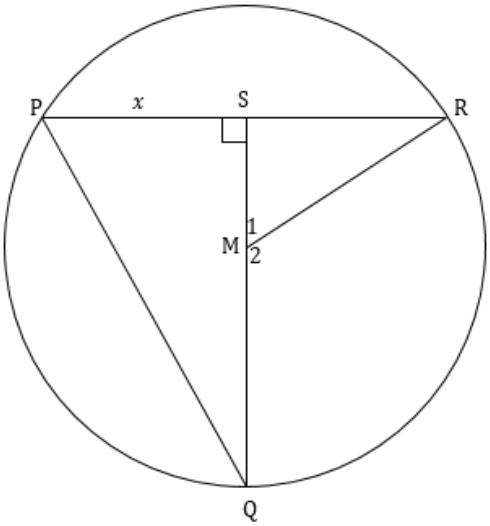
In the diagram, O is the centre of circle ABD. F is a point on chord AB such that DOF $\perp$ AB. AB $=$ FD $= 8$ cm and OF $= x$ cm.



Determine the length of the radius of the circle**.** **[5]**

Activity 6

In the diagram, PR and PQ are equal chords of the circle with centre M. QS is perpendicular to PR at S. PS $= x$ cm and MR is drawn.



6.1 Express, giving reasons, QS in terms of x. (5)

6.2 If $x = \sqrt{12}$ and MS $= 1$ unit, calculate the length of the radius of the circle. (2)

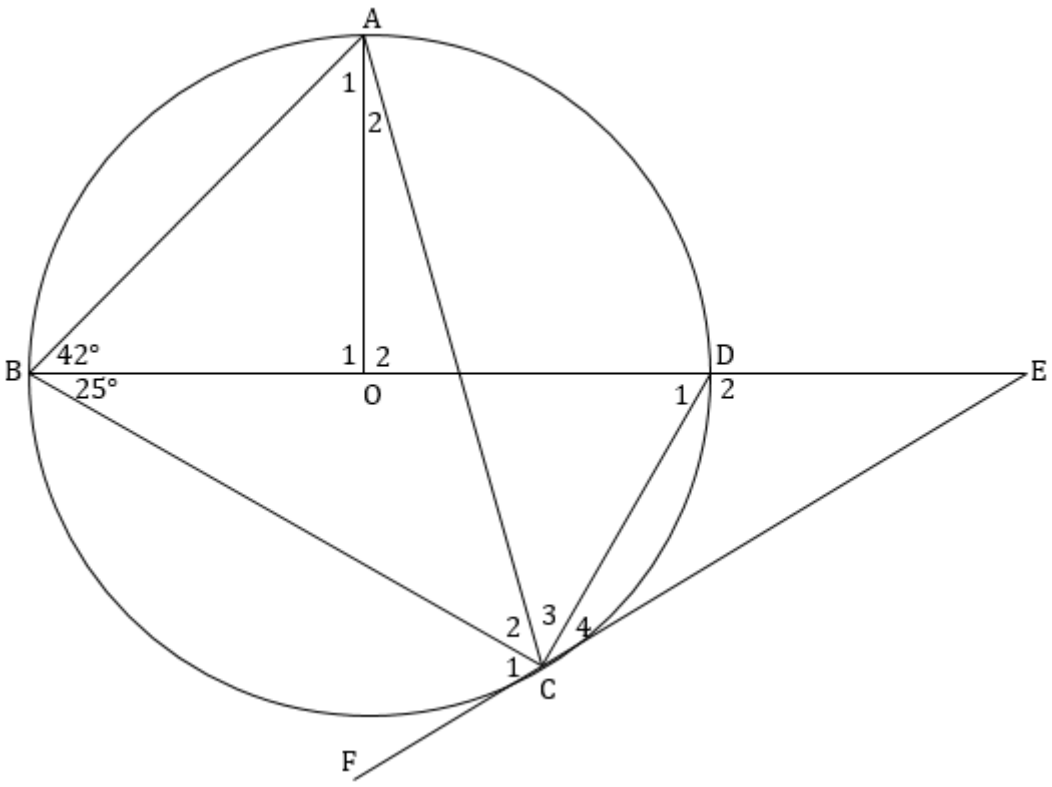
6.3 Calculate, giving reasons, the size of $\hat{P}$. (5)

[12]

TUTORIAL 2

Activity 1

In the diagram below, the circle with centre O passes through A, B, C and D such that BOD is a diameter. BD is extended to E such that FCE is a tangent to the circle at C. $A\hat{B}E = 42°$ and $D\hat{B}C = 25°$



Calculate:

1.1 $B\hat{C}D$ (2)

1.2 $\hat{A}_1$ (2)

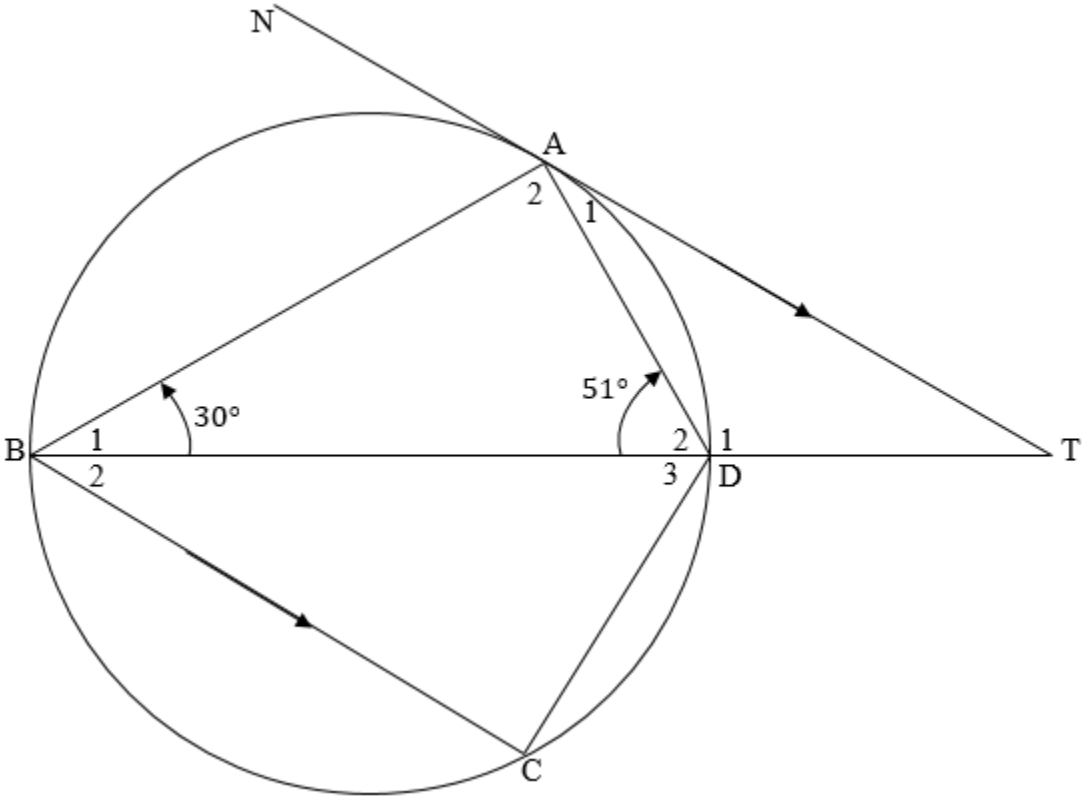
1.3 $\hat{O}_2$ (2)

1.4 $\hat{C}_4$ (2)

[8]

Activity 2

In the diagram, TAN is a tangent to the circle at A. ABCD is a cyclic quadrilateral. BD is drawn and produced to meet the tangent at T. $\hat{B}_1 = 30°$ and $\hat{D}_2 = 51°$. TAN||CB.



Calculate, giving reasons, the size of:

2.1 $\hat{A}_1$ (2)

2.2 $\hat{T}$ (2)

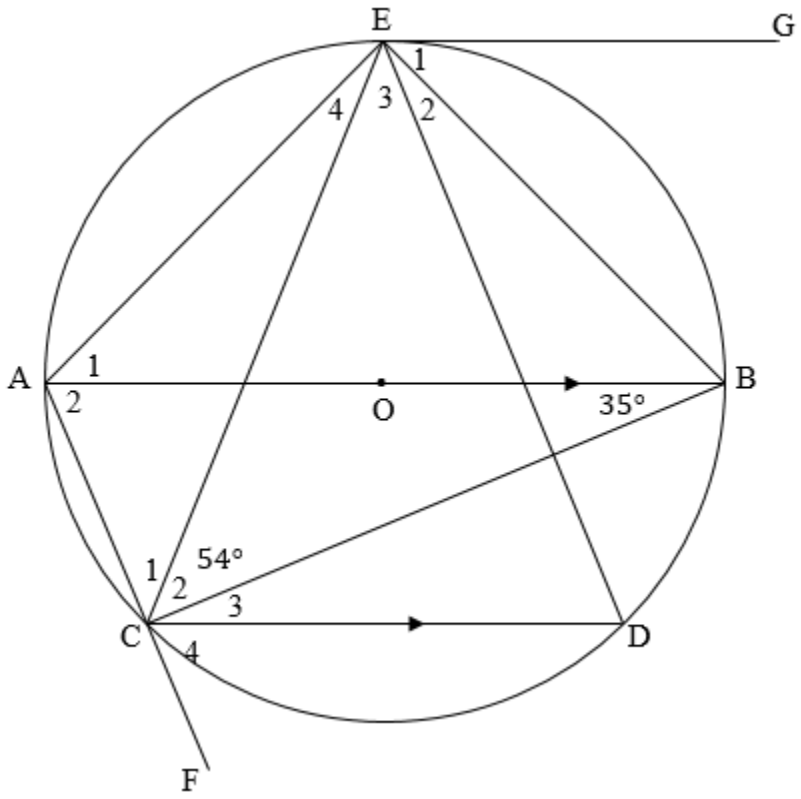
2.3 B_2 (2)

2.4 $\hat{C}$ (2)

[8]

Activity 3

O is the centre of the circle in the diagram with chord CD parallel to diameter AB. AC is produced to F and EG is a tangent to the circle. $A\hat{B}C = 35°$ and $\hat{C}_2 = 54°$.



Calculate, with reasons, the sizes of the following angles:

3.1 $\hat{E}_1$ (2)

3.2 $\hat{C}_1$ (2)

3.3 $\hat{C}_3$ (2)

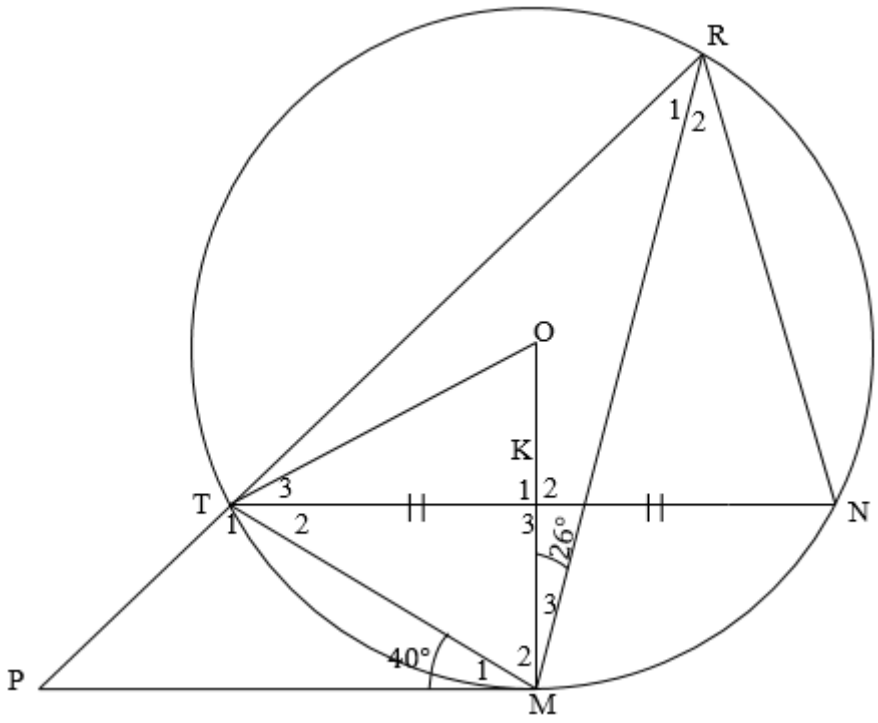
3.4 $A\hat{E}D$ (2)

3.5 $\hat{E}_3$ (3)

[11]

Activity 4

In the diagram below, O is the centre of circle TRNM. MP is a tangent to the circle at M such that RT produced meet MP at P. OM intersects TN at K. K is the midpoint of TN.
$P\widehat{M}T = 40°$ and $\widehat{M}_3 = 26°$.



Calculate, with reasons, the size of:

4.1 $T\widehat{O}M$ (4)

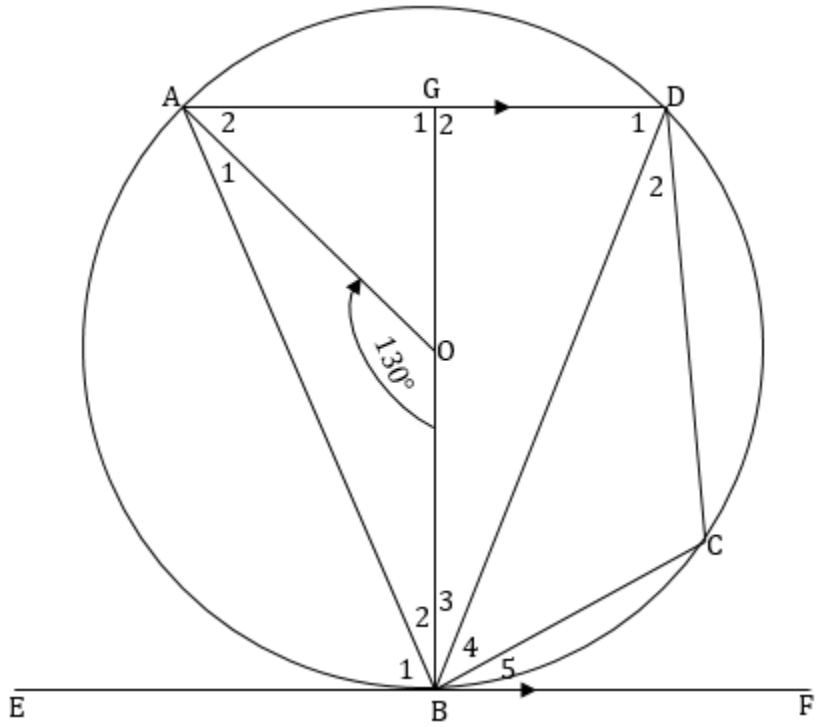
4.2 $\widehat{N}$ (4)

4.3 $\widehat{T}_3$ (3)

[11]

Activity 5

In the diagram below, the circle having centre O, passes through A, B, C and D, with $A\widehat{O}B = 130°$. EBF is a tangent to the circle at B with EF||AD. BOG is a straight line.



Calculate, with reasons, the size of:

5.1 $\widehat{D}_1$ (2)

5.2 $\widehat{B}_1$ (2)

5.3 $B\widehat{A}D$ (1)

5.4 $\widehat{C}$ (2)

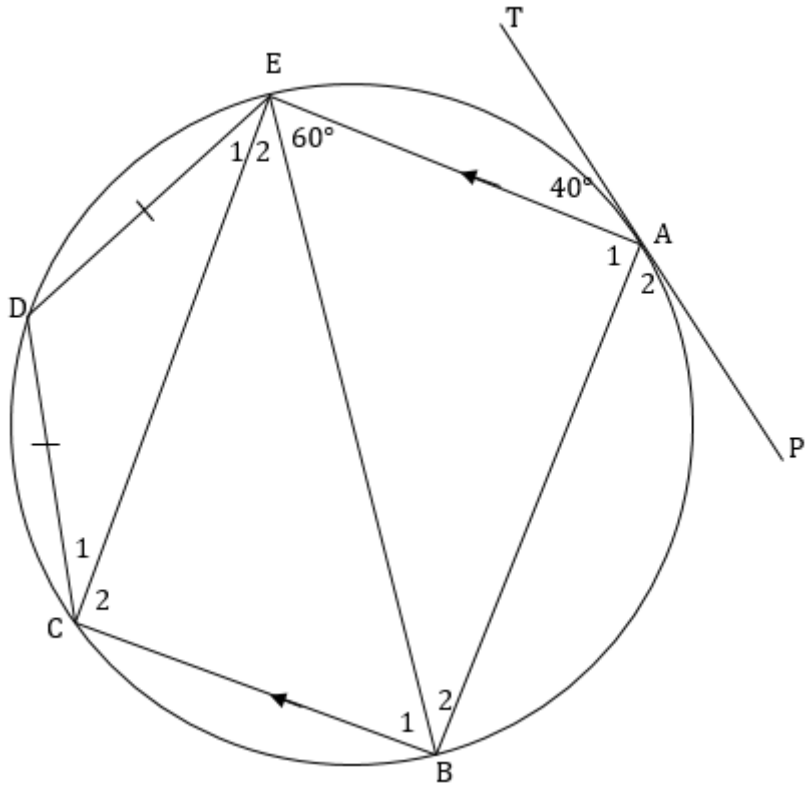
5.5 $\widehat{B}_3$ (3)

5.6 Calculate the length of GD, if AD $= \frac{\sqrt{7}}{2}$ units. (3)

[13]

Activity 6

In the diagram below, TAP is a tangent to circle ABCDE at A. AE||BC and DC = DE.
$\hat{TAE} = 40°$ and $\hat{AEB} = 60°$.



6.1 Identify TWO cyclic quadrilaterals. (2)

Calculate, with reasons, the size of:

6.2 $\hat{B}_2$ (2)

6.3 $\hat{B}_1$ (2)

6.4 $\hat{D}$ (2)

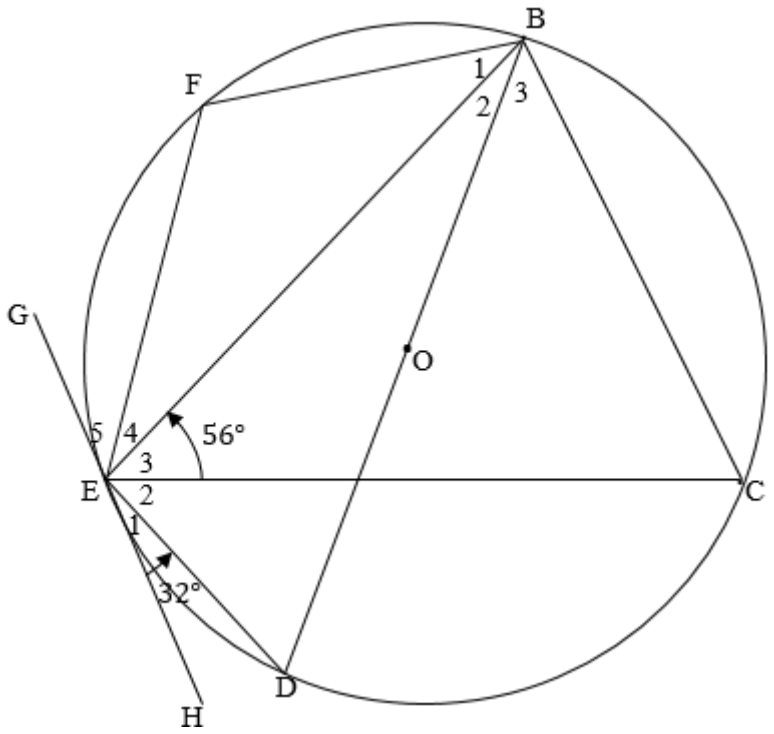
6.5 $\hat{E}_1$ (3)

[11]

TUTORIAL 3

Activity 1

In the diagram below, O is the centre of the circle. BD is a diameter of the circle. GEH is a tangent to the circle at E. F and C are two points on the circle, then FB, FE, BC, CE and BE are drawn. $\hat{E}_1 = 32°$ and $\hat{E}_3 = 56°$.



Calculate, with reasons, the size of:

1.1 $\hat{E}_2$ (2)

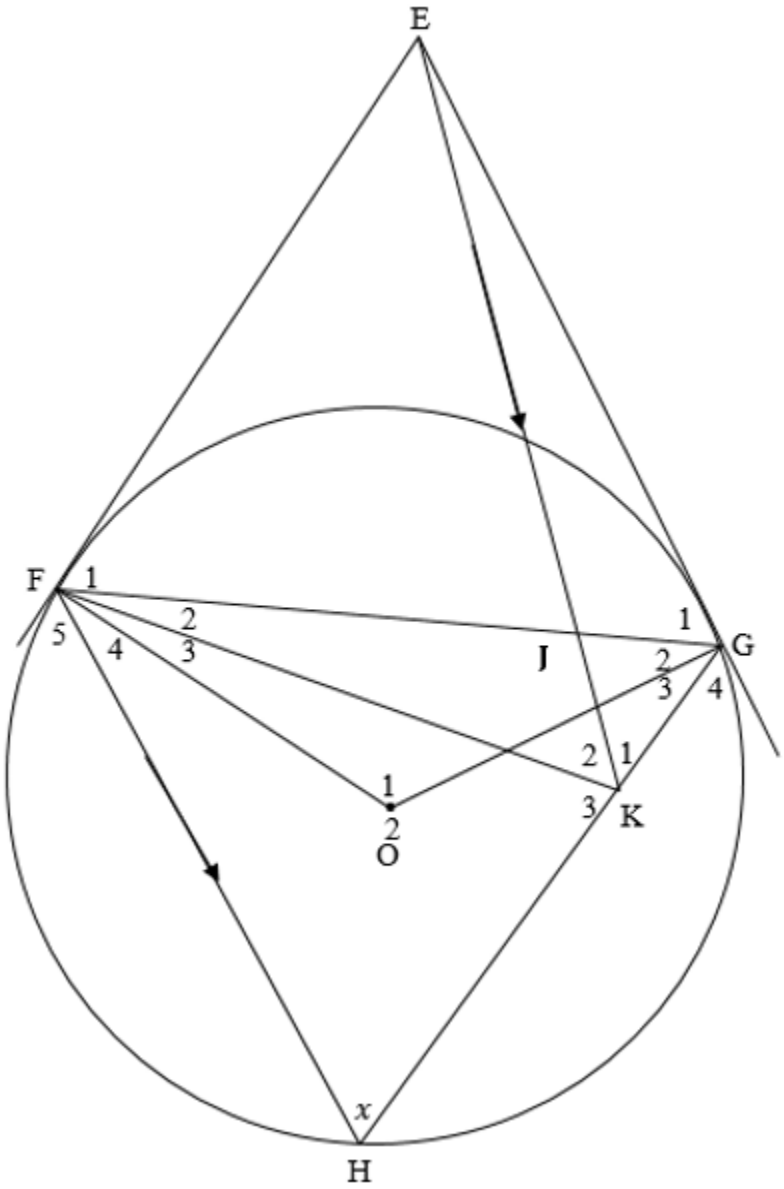
1.2 $E\hat{B}C$ (3)

1.3 $\hat{F}$ (4)

[9]

Activity 2

In the diagram, EF and EG are tangents to circle with centre O. FH||EK, EK intersects FG at J and meets GH at K. $\hat{H} = x$.



Prove that:

2.1 FOGE is a cyclic quadrilateral. (5)

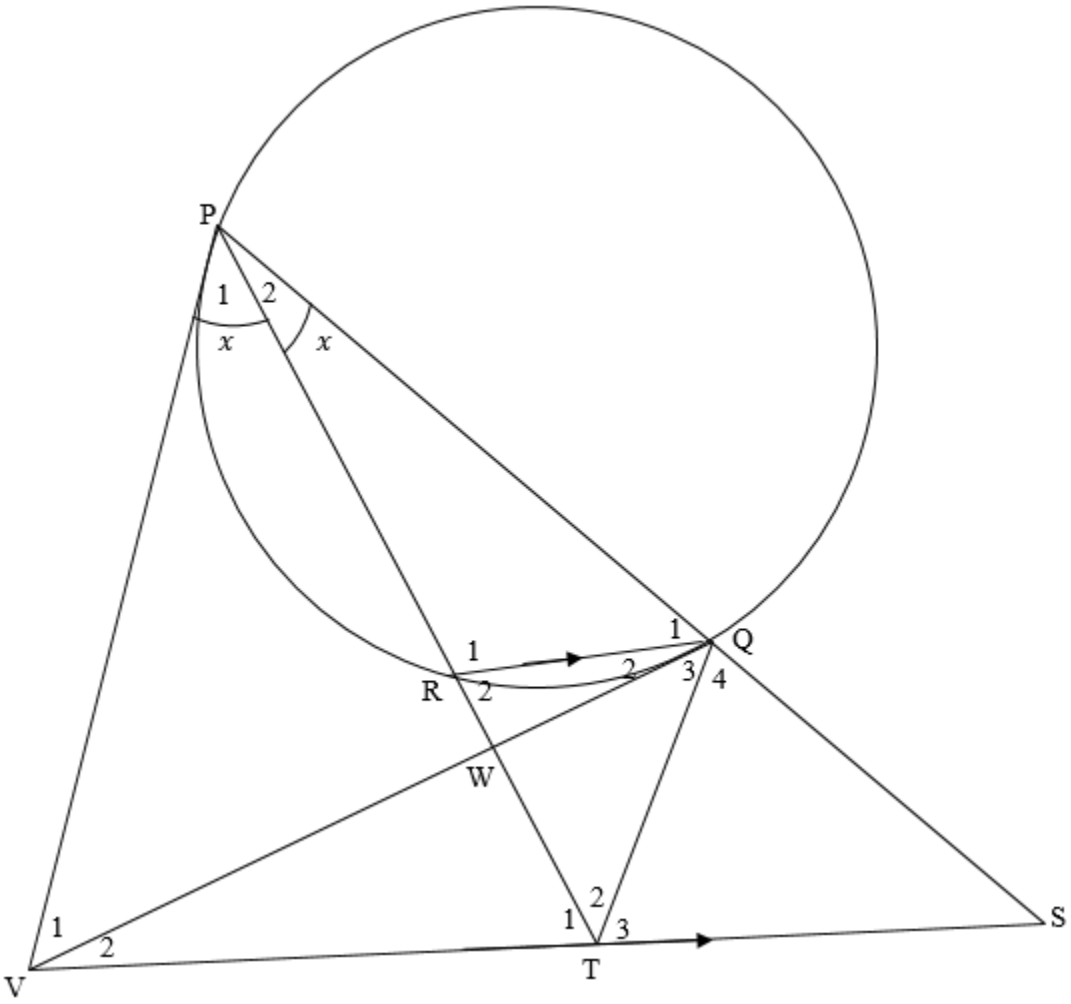
2.2 EG is a tangent to the circle GJK. (5)

2.3 $F\hat{E}G = 180° - 2x$. (3)

[13]

Activity 3

In the diagram, PV and VQ are tangents to the circle at P and Q. PQ is produced to S and chord PR is produced to T such that VTS||RQ. VQ and RT intersect at W. $\hat{P}_1 = \hat{P}_2 = x$.



Prove that:

3.1 $\hat{S} = x$ (4)

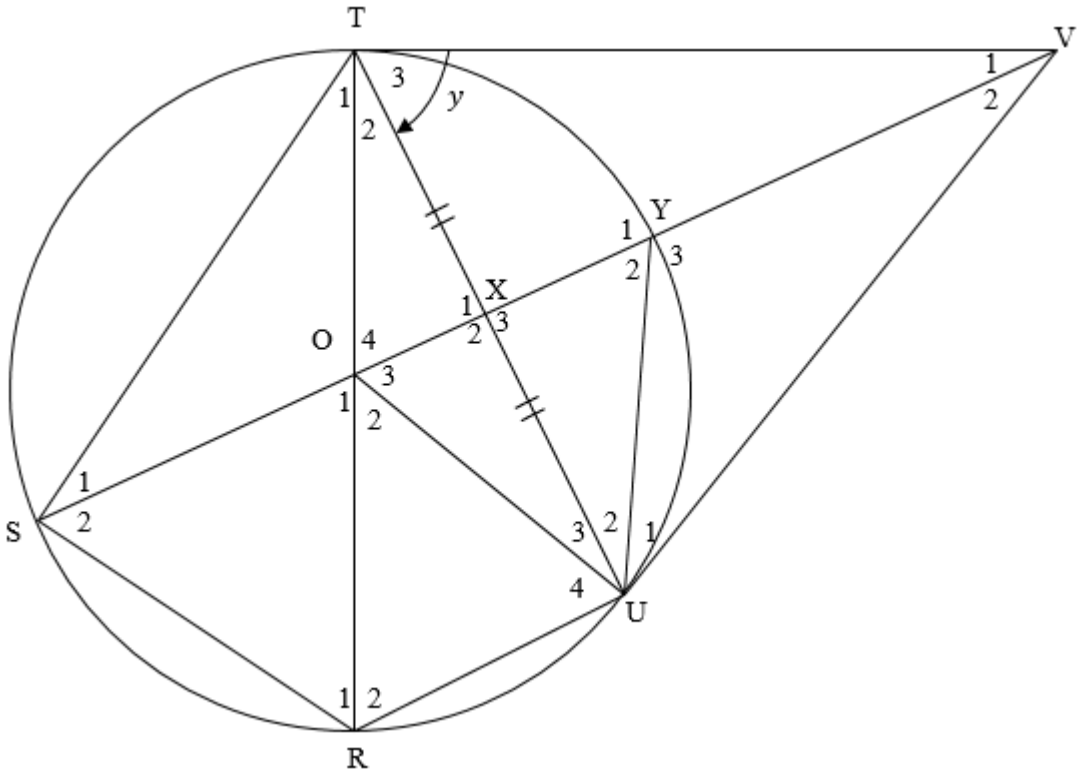
3.2 PQTV is acyclic quadrilateral. (5)

3.3 TQ is a tangent to the circle passing through Q, W and P. (3)

[12]

Activity 4

TV and VU are tangents to the circle with centre O at T and U respectively. TSRUY are points on the circle such chat RT is the diameter. X is the midpoint of chord TU. $\hat{T}_3 = y$.



Prove that:

4.1 RU||SY (5)

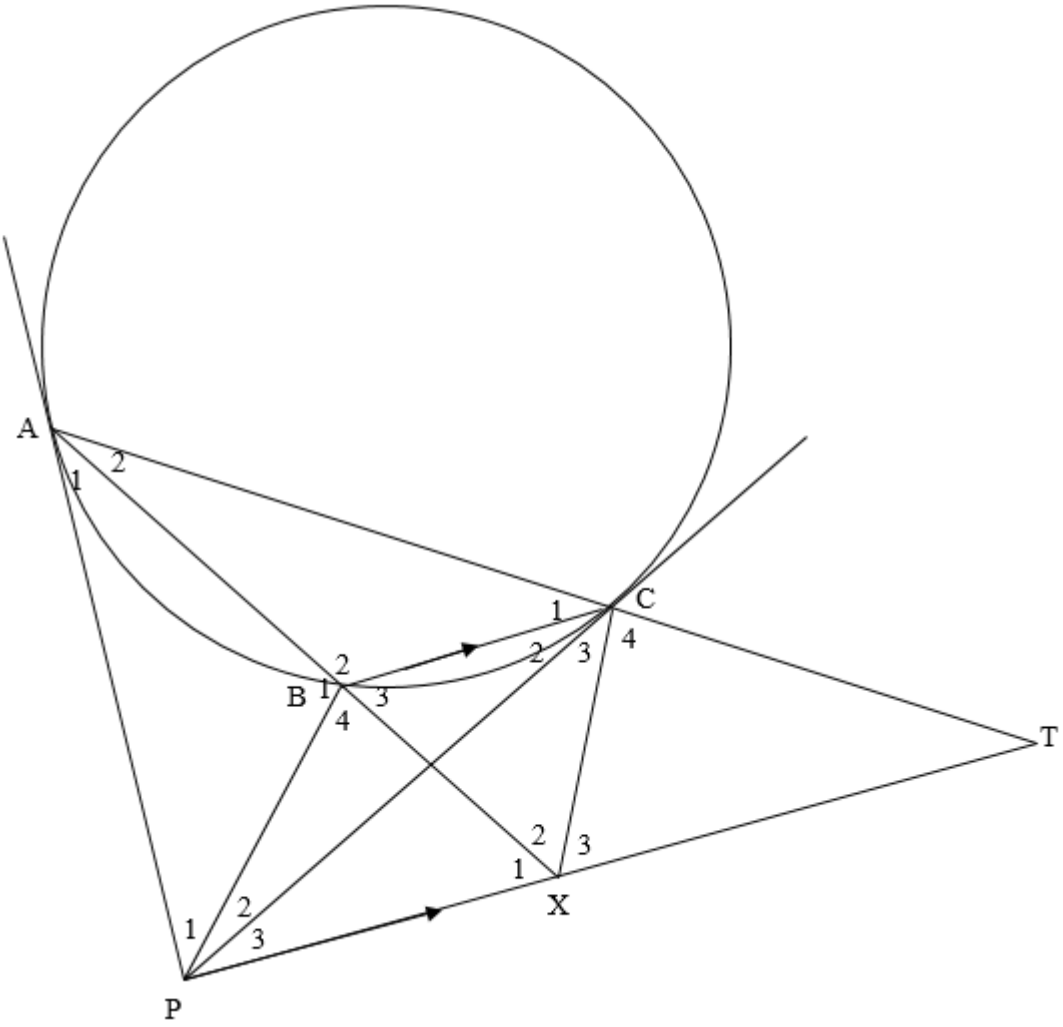
4.2 $\hat{T}_1 = \frac{1}{2}y$ (5)

4.3 TOUV is a cyclic quadrilateral (5)

[15]

Activity 5

In the diagram below, PA and PC are tangents drawn from point P to the circle ABC.
AB = BC. ABX is a straight line. AC and PX are produced to meet at T. BC||PT and BP, CP and CX are drawn.



Prove that:

5.1 AB bisects $P\hat{A}C$ (3)

5.2 BC bisects $P\hat{C}A$ (3)

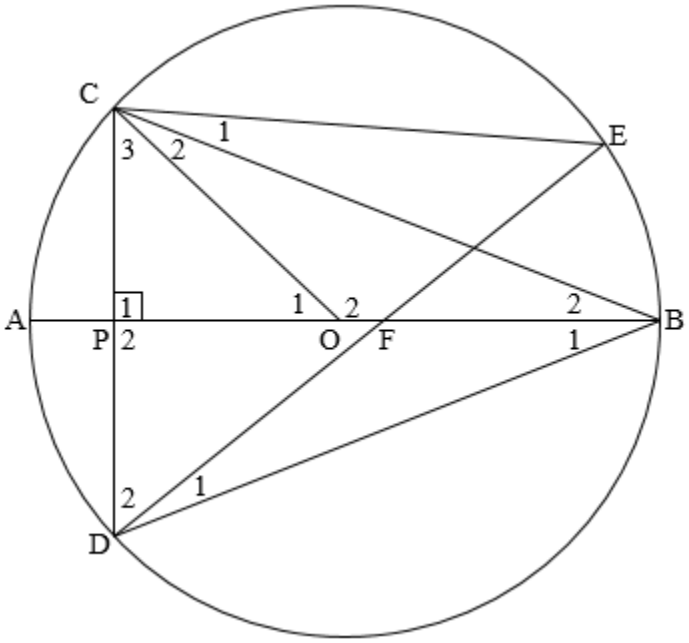
5.3 $P\hat{A}C = \hat{X}_3$ (4)

5.4 BC is a tangent to the circle CPT (4)

[14]

Activity 6

In the diagram below, O is the centre of the circle CADBE with AB as diameter, CD ⊥ AB and cuts point P. Chord DE cuts AB at F. CE, CB, CO and DB are drawn.



Prove that:

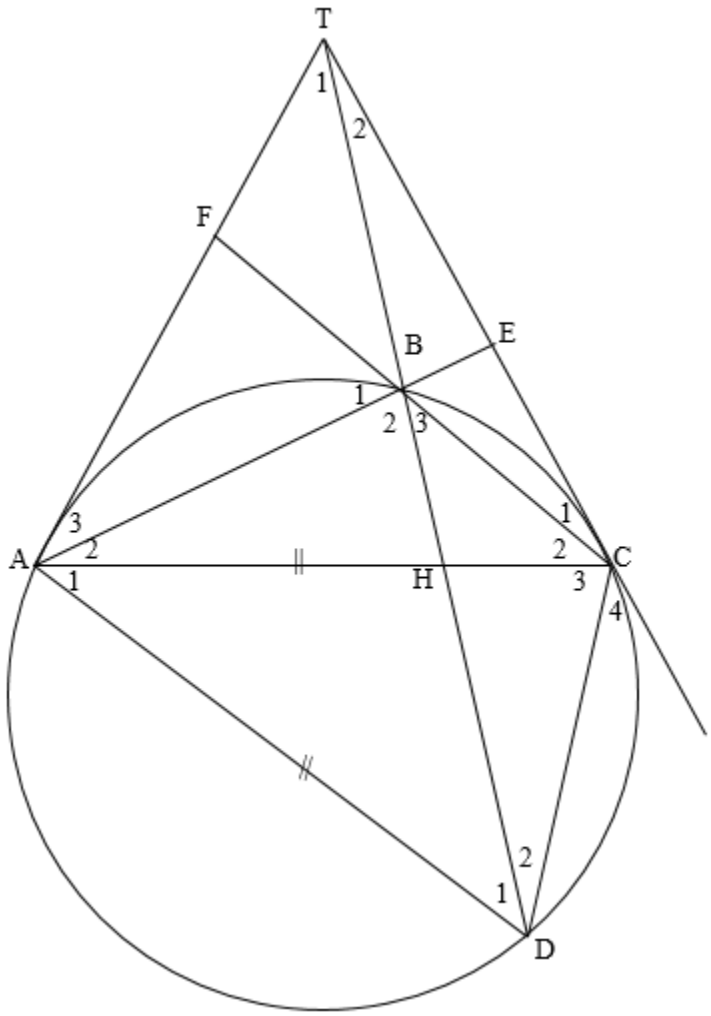
6.1 $\widehat{B}_1 = \widehat{B}_2$ (4)

6.2 CEFO is a cyclic quadrilateral (5)

[9]

Activity 7

In the diagram below, ABCD is a cyclic quadrilateral with AC = AD. Tangents AT and CT touch the circle at A and C respectively. FBC, ABE, AHC and DBT are straight lines.



7.1 $\widehat{B}_1 = \widehat{B}_2$ (5)

7.2 BECH is a cyclic quadrilateral (4)

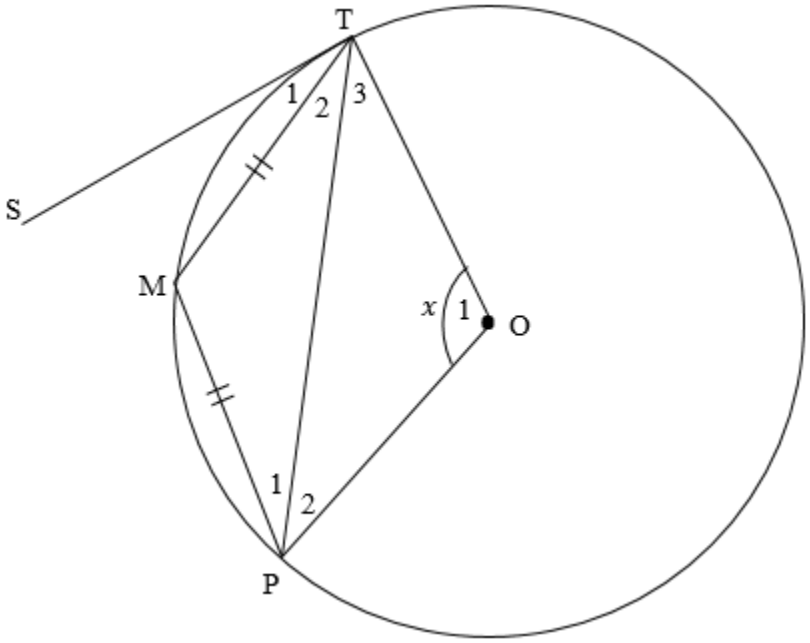
7.3 CA is a tangent to the circle passing through points A, B and T. (5)

[14]

TUTORIAL 4

Activity 1

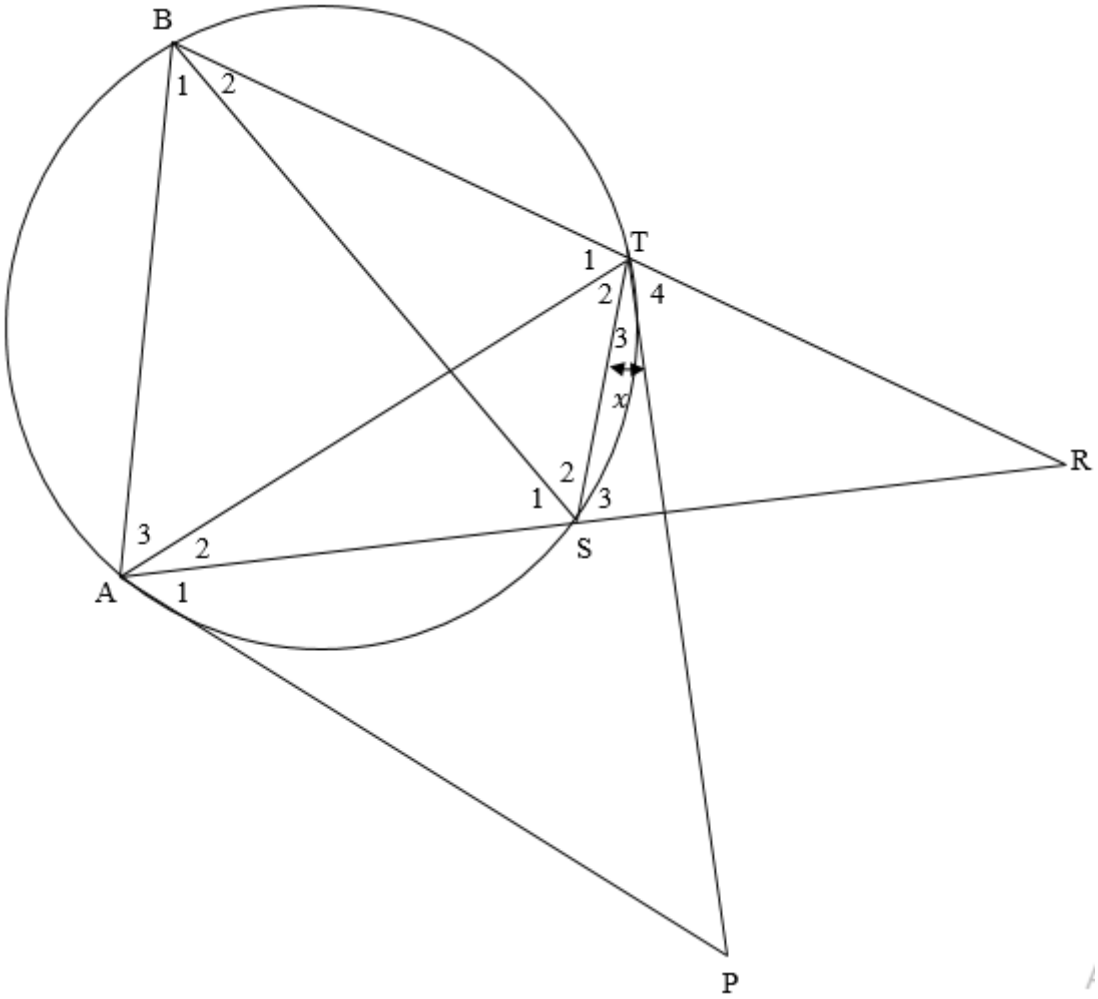
In the diagram, O is the centre of the circle. ST is a tangent to the circle at T. M and P are points on the circle such that TM = MP. OT, OP and TP are drawn. $\widehat{O}_1 = x$.



1.1 Prove that $S\widehat{T}M = \frac{1}{4}x$. **[7]**

Activity 2

In the diagram, PA and PT are tangents to a circle at A and T respectively. B and S are points on the circle such that BT produced and AS produced meet at R and BR = AR. BS, AT and TS are drawn. $\hat{T}_3 = x$.



2.1 Give a reason why $\hat{T}_3 = \hat{A}_2 = x$. (1)

2.2 Prove that AB||ST. (5)

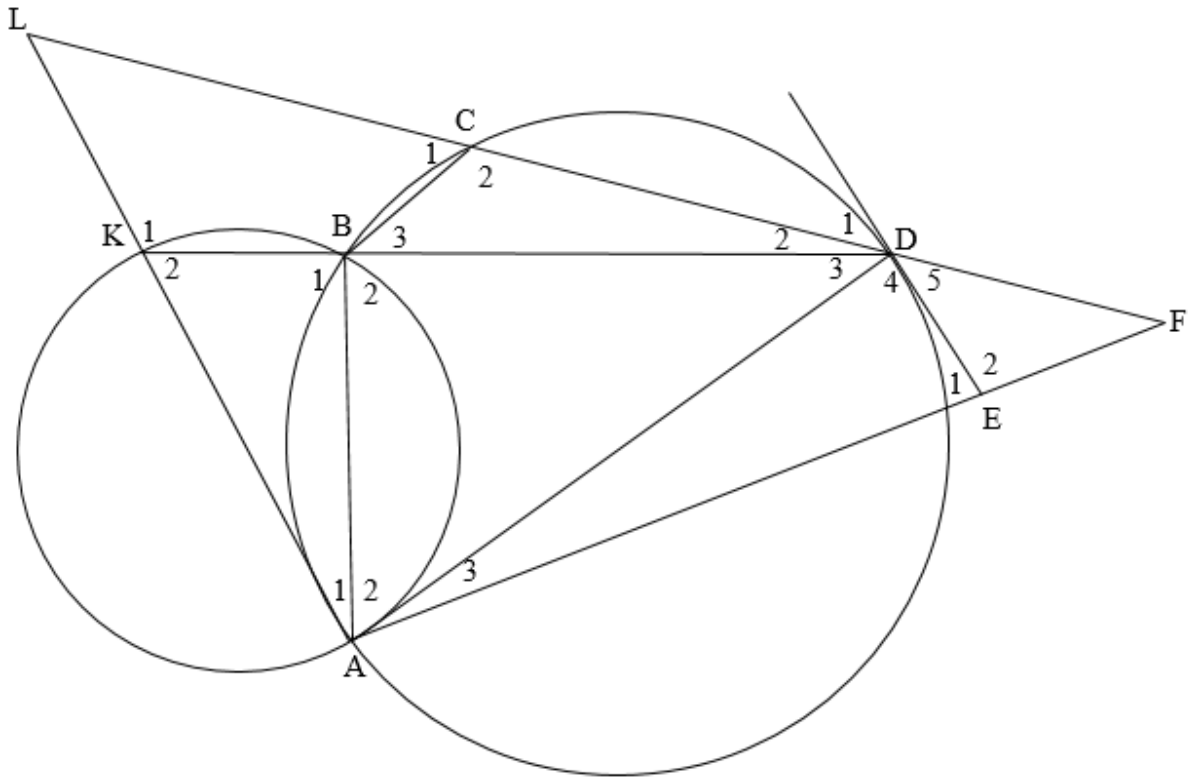
2.3 Prove that $\hat{T}_4 = \hat{A}_1$ (5)

2.4 Prove that RTAP is a cyclic quadrilateral. (2)

[13]

Activity 3

In the diagram below, two circles intersect each other at A and B. ED is a tangent to circle ABCD. DA is a tangent to circle AKB. DBK is a straight line. AK and DC are produced to meet at L. LCD and AE are produced to meet at F. CD = DF.



Prove that:

3.1 LKBC is a cyclic quadrilateral. (5)

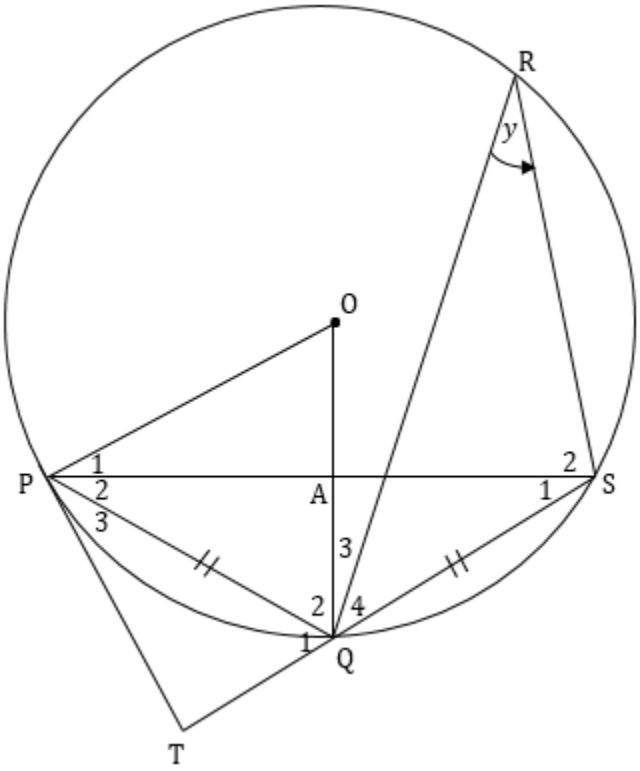
3.2 $\hat{B}_2 = L\hat{A}D$ (3)

3.3 DE||LA (5)

[13]

Activity 4

In the diagram, O is the centre of the circle and P, Q, S and R are points on the circle.
PQ = QS and $\hat{QRS} = y$. The tangent at P meets SQ produced at T. OQ intersects PS at A.



4.1 Give a reason why $\hat{P}_2 = y$ (1)

4.2 Prove that PQ bisects $T\hat{P}S$. (4)

4.3 Determine $P\hat{O}Q$ in terms of y. (2)

4.4 Prove that PT is a tangent to the circle that passes through P, O and A. (2)

4.5 Prove that $O\hat{A}P = 90°$. (5)

[14]